



Jvara

Learning Objectives

- ⇒ To know the exact meaning of Jvara
- ⇒ To do the Vyadhi Vinishchaya (diagnosis) of Jvara in proper order
- ⇒ To know the Vidhi Samprapti of Jvara and detailed explanation and current understanding wherever possible
- ⇒ To know and analyse the Nidana (aetiological factors) of Jvara, Samprapti (pathogenesis of Jvara), Samanya Purvarupa (commonly observed prodromal symptoms), Vishista Purvarupa (specific premonitory symptoms), Rupa (common clinical features), Pratyatma Lakshana (cardinal features) of Jvara
- ⇒ To describe Doshaja Jvara, Sannipataja Jvara
- ⇒ To describe Abhishanga, Vishama, Punaravartaka Jvara
- ⇒ To know about Antarvega and Bahirvega Jvara, Vatabalasaka Jvara, Pralepaka Jvara
- ⇒ To differentiate Ama, Pachyamana and Nirama Jvara
- ⇒ To analyse Doshaja and Agantuja Jvara with respect to Jvara Samprapti
- ⇒ To describe Sadhyasadyata of Jvara
- ⇒ To know the Vyadhyavastha (stage of the disease) in Jvara
- ⇒ To enlist the Jvara Mukta Lakshana

Introduction

Jvara (see Figure 39) is considered to be the first disease to evolve in the pathology as per *Ayurveda* literatures. This came into the existence due to the anger of Lord Rudra which was the outcome of departure of Goddess Sati in the event of *Daksha Yajna*. Jvara is the primary disease where all the creatures will definitely suffer at least once between life and death. It is characterised by increased temperature, malaise and loss of appetite. It can manifest as a primary disease or associated with other disease or even manifest as a complication. Presence of fever indicates that the disease is on progress and tend to worsen in due course of time. The course of the Jvara can be identified appropriately by the accurate and chronological clinical presentation in the ill individuals. The disease is considered to be *Pitta* dominant and *Amashayasamuttha Vikara* on account of *Pitta* dominance. It afflicts *Rasavaha Srotas* with *Sanga* and *Vimargagamana* as the type of *Srotodushti* happening throughout the course of the disease. *Svedavaha Srotas* will be afflicted

during the progression of the disease. Depending upon the nature of clinical presentation, *Srotas* other than *Rasavaha* and *Svedavaha Srotas* will be afflicted during the progression of the disease. For example, when an individual is presented with *Kasa* (cough) along with *Jvara* (raised body temperature), we may predict the affliction of *Pranavaha Srotas* along with the afflictions of the *Rasavaha* and *Svedavaha Srotas*. When an individual is presented with *Pita Tvak*, *Netra* (yellowish discolouration of the skin, sclera) and *Jvara* (raised body temperature), we may predict the involvement of *Raktavaha Srotas* along with *Rasavaha* and *Svedavaha Srotas*. Depending upon the age of the individual, duration of exposure to the aetiological factor, nature of severity of the aetiological factor, number of *Dosha* vitiated, number of *Dhatu* afflicted, number of *Srotas* involved, *Rogamarga* involvement throughout the progression of the disease etc determine the *Sadhyasadyata* of *Jvara*. In the following pages, we are diving deep into the subject to know more about this particular disease.

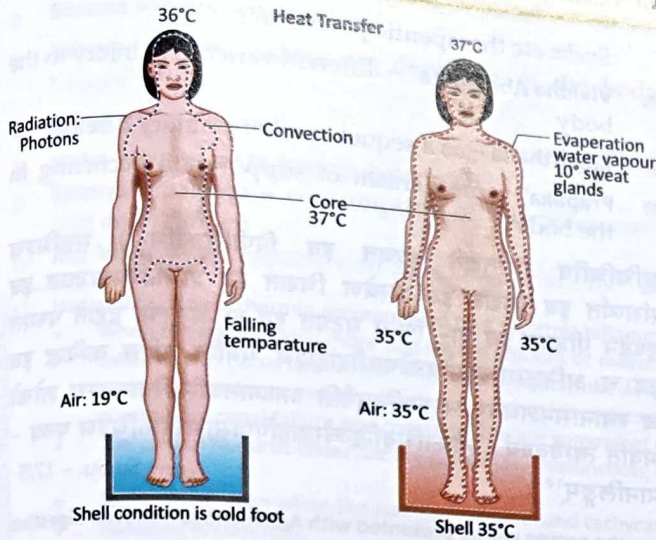


Figure 39: Shell and core temperature

देहेन्द्रियमनस्तापी सर्वरोगाग्रजो बली।

ज्वरः प्रधानो रोगाणामुक्तो भगवता पुरा॥४॥

Ca.Ci. - 3/4

Jvara is considered to be one of the very prime and important disease among all diseases. As there will be the affliction of *Deha* (physique), *Indriya* (sense organs and organs of action), *Manas* (mind or *Ubhayendriya*). In the coming pages, we are learning to identify the discomfort produced due to the affliction of *Deha*, *Indriya* and *Manas*.

Definition

ज्वरयति शरीराणीति ज्वरः,

Ca.Ni. – 1/35

ज्वरयति संतापयति।

Chakra, Ibid

ज्वरशब्देन च देहमनः संतापकरत्वं;

Chakra, Ca.Ni. – 1/5

सन्तापलक्षणो ज्वर

Aruna, A.Hr.Ni. - 2/3

ज्वरस्त्वेक एव संतापलक्षणः।

Ca.Ni. - 1/32

अत एव सर्वकायमन सन्तापकत्व ज्वरस्याहुः

Aruna, A.Hr.Ni. 1/1

सन्तापलक्षणो ज्वर

Dipika commentary, Sha.Pu. – 7/2

ज्वरप्रत्यात्मिकं लिङ्गं संतापो देहमानसः।

Ca.Ci. – 3/31

स्वेदावरोधः सन्तापः सर्वाङ्गग्रहणं तथा॥ १३ ॥

विकारा युगपद्वस्मिन् ज्वरः स परिकीर्तितः।

Su.Ut. – 39/13-14

By analysing above said descriptions available in different *Ayurvedic* literatures, it can be concluded that *Pitta Dosha* is the crucial *Dosha* responsible for the manifestation of *Jvara*. The following are the *Pratyatma* (cardinal) and essential features seen during the course of *Jvara*¹.

- ⊛ **Svedavarodha** – absence of sweating
- ⊛ **Santapa** of **Deha**² and **Manas**³ – discomfort related to physique and mind
- ⊛ **Sarvanga Grahana** – stiffness of all over the **body**⁴

Vidhi Samprapti⁵ or Classification of Jvara:

- ❖ **2 types:** *Sharirika* and *Manasika*
- ❖ **2 types:** *Saumiya* and *Agneya*
- ❖ **2 types:** *Antarvega* and *Bahirvega*
- ❖ **2 types:** *Prakrita* and *Vaikrita*
- ❖ **2 types:** *Sadhya* and *Asadhya*
- ❖ **5 types:** *Santata*, *Satata*, *Anyedyushka*, *Tritiyaka* and *Chaturthaka*
- ❖ **7 types:** *Rasadhatugata*, *Raktadhatugata*,
Mamsadhatugata, *Medodhatugata*, *Asthidhatugata*,
Majjadhatugata, *Shukradhatugata*

- 1 When one analyse the description available in the Lakshana of Jvara, we can observe the following pathological mechanisms occurring in the body.
- Peripheral vaso constriction (this is attributable to the absence of sweating)
 - Sarvanga Grahana (this is attributable to the shivering or chills) which occurs due to the muscle contraction which in turn generates heat
 - Raised body temperature and discomfort to the mind
- Either the increased heat production or hampered heat loss from the body are the two mechanisms leading to the increased temperature of the body.
- 2 Deha Santapa is identified by perception of increased body temperature either by the patient or the physician. This is best appreciated by keeping the dorsal surface of the palm either on the forehead or neck or on the skin of the suffering individual.
- 3 वैचित्यमस्तिग्लनिर्मनसस्तापलक्षणम् ।। Ca.Ci. – 3/36
- In Manasika Santapa, the person will be presented with abnormal functions of Chitta, restlessness and mental exhaustion.
- चित्त्यं विचार्यमूढं च ध्येयं संकल्प्यमेव च । यत्किञ्चिन्मनसो ज्ञेयं तत्सर्वं ह्यर्थसंज्ञकम् ।।
- इन्द्रियाभिग्रहः कर्म मनसः स्वस्य निग्रहः । ऊहो विचारश्च, ततः परं बुद्धिः प्रवर्तते ।। Ca.Sha. – 1/20,21
- In this situation when an individual is having abnormal functions of Manas, the person is unable to have appropriate Chintana, Vicharana, Uha, Dhyeya and Samkalpa or inappropriate thinking process or inappropriate predictability or inappropriate emotions and he is unable to concentrate the activities on any particular Indriya.
- 4 Stiffness all over the body parts implies the occurrence of muscle contraction which in turn leads to the increased heat production where in the person may present with chills or rigors in adults and convulsion or seizures in case of infants.
- 5 When Vidhi Samprapti of any disease is explained, it implies that various classification of the disease depending upon various parameters. For example, Sharirika and Manasika Jvara are the types based on the affliction of Ashraya of Vyadhi i.e. Sharira (physique) and Manasika (psyche). Each of this type of Jvara need different management approach. If management planned for a type of disease is not appropriate to that particular type of disease, the cure from that particular disease is remote. For example, if an individual is suffering from Manasika Jvara and no appropriate planning is being to take care of abnormal functioning of Manas, then success in attaining the relief from Manasika Jvara is almost and always remote.

- 8 types: Vataja, Pittaja, Kaphaja, Vata-pittaja, Pittakaphaja, Vata-kaphaja, Sannipataja, Agantuja

Nidana of Jvara

मिथ्यातियुक्तैरपि च स्नेहाद्यैः कर्मभिरनृणाम्॥११॥
विविधादभिघाताच्च रोगोत्थानात् प्रपातः।
श्रमात् क्षयादजीर्णाच्च विपात्सात्त्यंतुर्पर्ययात्॥१२०॥
ओषधीषुष्यगन्धाच्च शोकात्क्षत्रपीडनात्।
अभिचाराभिशापाभ्यां मनोभूताभिः॥१२१॥
स्त्रीणामपप्रजातानां प्रजातानां तथाहितैः।
स्तन्यावतरणे चैव ज्वरो दोषैः प्रवर्तते॥१२२॥

Su.Ut. - 39/19-22

- 6 Mithya and Atiyoga of Sneha⁶ etc Panchakarma (purificatory procedures) – improper administration of

When the person is administered with Sneha etc therapeutic procedures, the person will be presented with Agnimandya (reduced digestive fire). If he is exposed to the etiological factors, that will lead to the vitiation of Rasavaha Srotas

- The following verse describes various kinds of discomfort on non following of the appropriate dietary measures (during the state of Ajirna [indigestion] and Adhyashana [when the person consumes the food substances before the digestion of previously consumed Sneha]).

अजीर्णाध्यशनाभ्यां तु मुखशोषाध्मानशूलनिस्तोदपिपासागात्रसादच्छर्यतीसारमूर्च्छाज्वरप्रवाहणामविषादयः (5); Ca.Si. - 12/14-5
The clinical features observed on improper post-Shodhana measures are Mukhashosha (dryness of the mouth), Adhmana (distension of the abdomen), Shula (pricking type of pain), Nistoda (pricking type of pain), Pipasa (thirst), Gatrasada (sluggish movement of body parts), Char-di (vomiting), Atisara (loose stools), Murcha (altered state of consciousness), Jvara (fever), Pravahana (excess straining for defecation) and Amavisha (toxins formed in the Annavaha Srotas). If we analyse the clinical features, most of them are indicative of abnormalities happening in the Mahasrotas or alimentary canal.
Thus, improper pre-Shodhana, Shodhana and post-Shodhana measures may first impair the Agni which is followed by Ama formation which is followed by affliction of Rasa and thus afflicting Rasavaha Srotas.

Ca.Ni. - 1/30

- 7 तत्राभिघातजो वायुना दुष्टशोणिताधिष्ठानेन
The vitiation of Vayu along with impaired Shonita or Rakta Dhatu will be observed in this kind of Jvara.
In this situation, we can predict the possibility of entry of microorganisms to the injured parts which leads to infection of the wound resulting in the occurrence of fever. This can be attributable to the increased activity of WBCs against microorganisms entered through injured areas.
- 8 Jvara is also manifested due to the affliction of certain diseases such as Rajayakshma or Raktapitta or Pittaja Kasa or Vidradhi etc where dominance of Pitta can be observed.
Any condition which is leading to either increased heat production or decreased heat loss from the body will lead to the manifestation of fever.

When we speak of heat production, the following factors will lead to increased heat production.

- Basal rate of metabolism of all cells of the body
- Extra rate of metabolism caused by muscular activity including muscle contraction and shivering
- Extra metabolism caused by the effect of thyroxin and other hormones such as growth hormone and testosterone
- Extra metabolism caused by the effect of epinephrine, norepinephrine and sympathetic activity on cells
- Extra metabolism caused by increased chemical activity in the cells themselves
- Extra metabolism needed for the digestion, absorption and storage of food.

When we speak of prevention of heat loss can occur because of generalised peripheral vasoconstriction or cold environment.

- 9 When neutrophils and macrophages engulf large number of bacteria and necrotic tissue, essentially all neutrophils and many, and most of the macrophages will die at the area of pus formation. After several days, a cavity is often excavated in the inflamed tissues that contains varying portions of necrotic tissue, dead neutrophils and dead macrophages and tissue fluid. This mixture is known as pus. After the infection has been suppressed, the dead cells and necrotic tissue in the pus gradually autolyse over a period of days, and the end products are eventually absorbed into the surrounding tissues and lymph until most of the evidence of tissue damage is gone.
- When mechanism of suppuration is happening, the following features are seen: pricking type of pain, sensation of bite of ants, sensation of movement of ants all over the body, sensation of injury by weapons, discomfort similar to the discomfort caused by beating with the sticks, burn injury, discomfort similar to scorpion bite, discolouration of the skin, increase of swelling, raised body temperature, burning sensation, thirst, tastelessness.

- 10 Pricking type of pain as if multiple needles are used to prick the skin; tingling sensation of the body parts as if ants are moving across the body; tearing type of pain as if sharp instruments are used to tear the body parts; piercing type of pain; pain in the body parts similar to the pain produced by the beating by the stick; squeezing type of pain with palm or fingers or burning type of pain, pain similar to the discomfort produced by pus formation in the body or due to the application of cautery, alkali or burning sensation in one part of the body or all parts of the body; severe pain as if bitten by scorpion; swelling; discolouration of the skin, worsening of the swelling, raised body temperature, burning sensation, thirst, tastelessness are the clinical features observed in Pachyamana Shotha.

Sneha etc therapeutic procedures.
Vividha Abhigata⁷ – different varieties of injury to the body

- Rogotthana – as a sequel to other primary disease⁸
- Prapaka⁹ – mechanism of suppuration occurring in the body

मुचिभिरिव निस्तुद्यते दृश्यत इव पिपीलिकाभिः स्तापिच संसर्प्यत इव छिद्यत इव शस्त्रेण भिद्यत इव शक्तिभिस्ताड्यत इव दण्डेन पीड्यत इव पा- पिना घट्यत इव च अङ्गुल्या दह्यते पच्यत इव च अनिक्षाराभ्यामोषचोषपरीदाहाश्च भवन्ति वृश्चि कविद्ध इव च स्थानासनशयनेषु न शान्तिमुपैति आध्मातबस्तिरिवाततश्च शोफो भवति त्वगवैवर्ण्यं (शोफाभिवृद्धिर्ज्वरदाहपिपासाभक्तारुचिश्च पच्यमानलिङ्गम्¹⁰);

Su.Su. - 17/5

- ✱ **Shrama** – excessive exertion¹¹
- ✱ **Kshaya** – *Dhatu Kshaya* i.e. depletion of the body tissues¹²
- ✱ **Ajirna** – indigestion¹³
- ✱ **Visha** – exposure to toxic substances¹⁴
- ✱ **Satmya Viparyaya** – consumption of unwholesome¹⁵ diet and regimen
- ✱ **Rtu Viparyaya** – exposure to seasonal variations¹⁶ or

- bad climate such as exposure to cold breeze in winter or increased time of exposure cold in cold environment or swimming in extreme cold water etc
- ✱ **Oshadhi Pushpa Gandha** – inhalation¹⁷ of pollens or fragrance of flowers of such plants
- ✱ **Shoka** – grief¹⁸
- ✱ **Nakshatra Pida** – due to the influence of cosmic force emerged from placement of stars

- 11 Heatstroke is hyperthermia accompanied by a systemic inflammatory response causing multiple organ dysfunction and often death.
 - When the person does more and severe amount of exercise, the person may develop extreme raise of temperature. This condition is called as exertional heatstroke. This type of heatstroke develops abruptly in healthy active people (e.g. athletes, military recruits, factory workers). Intense exertion in a hot environment causes a sudden massive heat load that the body cannot modulate. Rhabdomyolysis is common; renal failure and coagulopathy are somewhat more likely and severe.
 - The clinical features observed are global CNS dysfunction, ranging from confusion to delirium, seizures and coma, are the hallmarks of heatstroke.
 - Tachycardia, even when the patient is supine, and tachycardia are common.
 - Sweating is common.
 - Diagnosis is thorough clinical evaluation, including core temperature measurement and laboratory testing for organ dysfunction.
- 12 Raised body temperature can occur because of two mechanisms either due to excess generation of the heat (hot environment with humidity and excess metabolism) and reduced heat loss from the body (radiation, evaporation and convection).
 - Excess metabolism will alter the physiological functions leading to dehydration.
 - Whenever there is raised body temperature, this will lead to increase in peripheral circulation leading excess production of sweat. Normal cutaneous blood flow occurs around 200 to 250 ml/ min at normal temperatures, will increase to 7 to 8 Litres/ min with heat stress, requiring increase in cardiac output. Sweat production from negligible to 2 Litres/ hour, thus, the person will develop rapid dehydration.
- 13 After a meal ingested, the metabolic rate increases as a result of the different chemical reactions associated with digestion, absorption and storage of food in the body. This is called thermogenic effect of the body, because these processes require energy and generate heat.
 - After a meal that contains large quantity of carbohydrates or fats, the metabolic rate increases about 4%. However, after a high protein meal, the metabolic rate begins raising within an hour, and this lasts for 3 to 12 hours. This is responsible for the small amount of rise of temperature in the body.
 - When the person is suffering from indigestion, the body tries to get the nutrition through alternate sources. This is responsible for the alternate source of heat production i.e. skeletal muscle contraction which is usually responsible for the chills and rigors. Skeletal muscle stimulation, normally accounts for 20-30% of the basal metabolic rate. In case of fever, basal metabolic rate will increase.
- 14 Nature of toxic substances and exposure of the part of the body exposed to toxic substances decide the nature of clinical presentation. Certain drugs such as caffeine, cocaine, amphetamines are leading to the manifestation of fever. When an individual is exposed to toxic substances, there will be release of histamine released which is leading to the vasodilatation of the exposed area which in turn leads to increased permeability of the capillaries. As there is increased vascularity in the exposed area, the person may have local raise of body temperature.
- 15 This refers to the unwholesome diet pattern. Let us take an example, if a South Indian person going to northern part of India, he will have difficulty in consuming substances prepared there. Most of the dishes of North India will have mustard oil and South Indian will have habituation of consumption of food prepared out of coconut oil. That is why, the person from South India finds difficulty in digesting the food which is prepared from mustard oil. This may release the histamine which is leading to the local permeability of the blood to increase in the alimentary canal, stomach in particular.
- 16 Seasonal variations are of two categories: extreme hot environment and extreme cold environment. Apart from these two extremes, the variable environmental conditions in between seasonal extremes may be present. In case of extreme hot environment, excess heat is not disposed to the outside as the heat in the outside environment is extremely high; in case of extreme cold environment, heat is not expelled out as the environment is very cold. Abnormal variations may lead to variable loss of heat from the body, thus the person may have raised body temperature in any of the above mentioned situations.
- 17 This may lead to the respiratory allergic manifestation leading to inflammatory changes and lead to raise of temperature which is one of the essential sign of inflammation.
- 18 When an individual is in grief, the first and foremost effect is seen in Amygdala which is further controlled by hypothalamus. The effects of amygdala on the body are
 - Increase or decrease in heart rate
 - Increase or decrease in gastrointestinal secretion and motility
 - Defaecation or micturition
 - Pupillary dilatation
 - Piloerection
 - Secretion of various anterior pituitary hormones
 Increased cellular activity will lead to increased heat production and decreased cellular activity may lead to decreased heat loss from the body. In both occasions, the person may have feverishness.

- ⊗ **Abhichara** – bad deeds¹⁹
- ⊗ **Abhishapa** – curse or wrath by elders or teachers²⁰
- ⊗ **Manobhutabhishankaya** – people who are psychologically weak individuals
- ⊗ **Strinam Apaprajatanam and Prajatanam Ahita Bhojana** – improper delivery²¹ in pregnant women and in women indulging in improper diet²² and regimen²³ who gave normal birth to healthy baby
- ⊗ **Stanyavarana**²⁴ – in post natal period, when proper aseptic precautions are not taken during breast feeding

Samprapti²⁵ of Jvara

दोषाः प्रकुपिताः खेषु कालेषु स्वैः प्रकोपणैः॥१५॥

व्याप्य देहमशेषेण ज्वरमापादयन्ति हि।

दुष्टाः खहेतुभिर्दोषाः प्राप्यामाशयमृष्मणा॥१६॥

सहिताश्च समागत्य रसखेदप्रवाहिणाम्।

स्रोतसां मार्गमावृत्य मन्दीकृत्य हुताशनम्॥१७॥

निरस्य बहिरुपमाणं पक्तिस्थानाच्च केवलम्।

शरीरं समभिव्याप्य खकालेषु ज्वरागमम्॥१८॥

जनयन्त्यथ वृद्धिं च स्ववर्णं च त्वगादिषु।

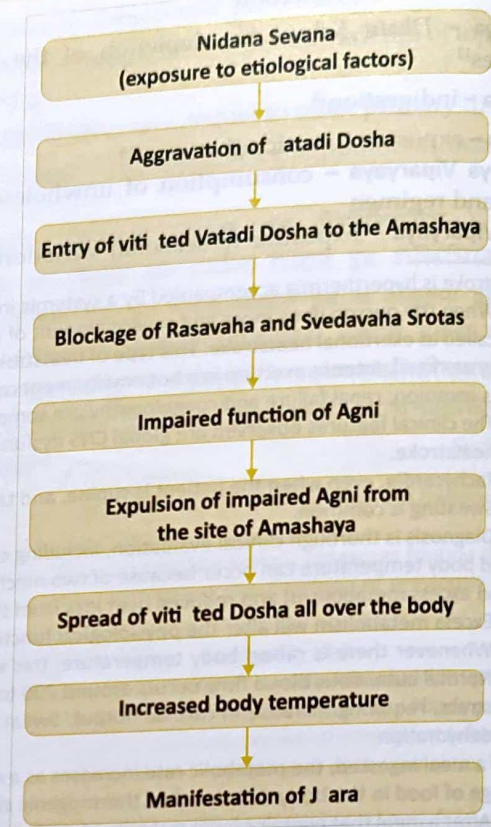
Su.Ut. – 39/15-19

मिथ्याहारविहारभ्यां दोषा ह्यामाशयाश्रयाः।

वह्निर्निरस्य कोष्ठानि ज्वरदाः स्यू रसानुगाः॥१९॥

Ma.Ni. – 2/2

- ⊗ Due to the exposure to either single or multiple aetiological factors, *Dosha* gets vitiated in *Amashaya*
- ⊗ Formation of *Ama* in the *Koshtha*
- ⊗ This *Ama* will lead to the displacement of *Agni* from *Koshtha* to *Shakha*
- ⊗ *Ama* will block the opening of *Rasavaha Srotas* so that *Agni* is not returned to *Koshtha*.
- ⊗ Due to displacement of *Pitta* from *Koshtha* to *Shakha*, there will be increased temperature felt in patient by touching the skin of the patient.
- ⊗ Due to the absence of *Agni* in *Mahasrotas*, the person will be presented with loss of appetite and indigestion.



Samprapti Ghataka

- ⊗ **Dosha** : *Pitta* dominant *Tridosha*
- ⊗ **Dushya** : *Rasa*
- ⊗ **Srotas** : *Rasavaha Srotas*
- ⊗ **Srotodushti Prakara** : *Sanga* and *Vimargagamana*
- ⊗ **Udbhava Sthana** : *Amashaya*
- ⊗ **Sanchara Sthana** : *Sarva Sharira*
- ⊗ **Vyakta Sthana** : *Tvak* or *Shakha*
- ⊗ **Adhishthana** : *Sharirika*

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- 19 This is responsible for the manifestation of Abhicharaja Jvara.
- 20 This is responsible for the manifestation of Abhicharaja Jvara.
- 21 This can be attributable to the improper wound care or attributable to abnormal measures followed during puerperium.
- 22 Consumption of black gram items etc may lead to Kapha vitiation which in turn lead to the formation of pus etc. Black gram (*Masha*) is having *Madhura Rasa*, *Guru*, *Snigdha* quality, *Shita Virya* and *Madhura Vipaka*, thus the person will have maximum Kapha *Dosha* vitiation.
- 23 Measures which lead to the increase in intrabdominal pressure such as squatting posture or excess straining for defaecation or vomiting or coughing etc
- 24 When the woman is in child bearing age group, particularly during lactation or there is inadequate treatment of acute mastitis, the lady may present with pain in the breast, throbbing type of pain, fever. This may occur due to entry of bacteria either through blood stream or through lactiferous ducts.
- 25 When an individual is having inappropriate diet pattern, that will cause the accumulation of improper digested materials in the *Amashaya* (stomach). This will be in the macromolecular or intermediate molecular form. This will lead to the infective and inflammatory changes in the stomach. This will be responsible for gastric mucosal damage. Because of the damage to gastric mucosa, macromolecular or intermediate molecules will get into blood circulation either via portal circulation or via lymphatics, macromolecules and intermediate molecules will enter the blood stream. The whole mechanism will lead to poor gastrointestinal function in due course of time.
 - Whenever, the person is suffering from *Jvara*, first and foremost, the blood circulation to the alimentary canal is going to get decreased. This in turn will lead to the increased blood circulation to the area of pathological events taking place inside the human body either it may be intravascular or interstitial or intracellular compartment.

- ❖ **Agni** : *Jatharagnimandya*
- ❖ **Ama** : Ama due to poor functioning of *Jatharagni*
- ❖ **Rogamarga** : *Abhyantara* and *Bahya*
- ❖ **Roga Svabhava** : *Ashukari*

Purvarupa

श्रमोऽरतिर्विवर्णत्वं वैरस्यं नयनप्लवः।
 इच्छाद्वेषौ मुहुश्चापि शीतवातातपादिषु॥25॥
 जृम्भाङ्गमर्दो गुरुता रोमहर्षोऽरुचिस्तमः।
 अहर्षश्च शीतं च भवत्युत्पत्स्यति ज्वरे॥26॥
 सामान्यतो विशेषात्तु जृम्भाऽत्यर्थं समीरणात्।
 पित्तान्नयनयोर्दाहः कफान्नाभिनन्दनम्॥27॥
 आलस्यं नयने साले जृम्भणं गौरवं क्रमः।
 ज्वलनातपवाय्वम्बुभक्तिद्वेषावनिश्चितौ॥28॥

Su.Ut. – 39/25-27

अविपाकास्यवैरस्ये हानिश्च बलवर्णयोः।
 शीलवैकृतमल्पं च ज्वरलक्षणमग्रजम्॥29॥

Ca.Ci. – 3/28-29

- ❖ *Shrama*²⁶: fatigue by doing activities
- ❖ *Arati*²⁷: restlessness
- ❖ *Vivarnatva*²⁸: discolouration
- ❖ *Vairasya*²⁹: abnormal taste perception
- ❖ *Nayana Plava*³⁰: tear production
- ❖ *Shita Vata Atapadishu Muhu Iccha Dvesha*³¹: frequent liking and disliking towards cold wind and sunlight
- ❖ *Jrimbha*³²: excess yawning
- ❖ *Angamarda*³³: bodyache
- ❖ *Guruta*³⁴: heaviness of the body
- ❖ *Romaharsha*³⁵: horripilation

- 26 This presenting complaint is attributable to poor blood circulation to the skeletal muscles secondary to peripheral vaso constriction. This will result in either excess carbonic acid production or lactic acid formation which are responsible for muscular fatigue.
- 27 This is attributable to hyperpyrexia where the temperature is more than 104° C or when the person is showing rapid development of fever. This may be associated with chills or rigor.
- 28 वर्ण मेदेन ग्लानिहर्षरौक्ष्यस्नेहा व्याख्याताः ।।
 The term Varna refers not only related to discolouration, it also encompasses abnormal texture of the skin too. Hence, Vaivarnya can be considered as not only abnormal discolouration, it also covers abnormal texture of the skin such as dryness of the skin or oedema of the skin or subcutaneous tissue etc.
 In the present context, Vaivarnya is attributable to abnormal increase of blood circulation to the affected part which is leading to redness of the body parts, thereby leading to the increase of local rise of temperature. Depending upon the organ afflicted, the person may have yellowish discolouration (hepatitis) or abnormal oedema formation (local inflammation) or skin turgor loss secondary to extreme fluid loss (hypovolemia) etc.
- 29 This is attributable to activation of sympathetic nervous system which in turn reduces the secretions in the whole alimentary canal. In this occasion, there will be poor salivary secretions which is essential for the proper taste perception.
- 30 This is attributable to the abnormal respiratory function. In this occasion, the person will have blockage of the nasolacrimal duct, thus there will be reverse movement of lachrymal secretions, thus the patient will have excess tear production.
- 31 This is attributable to the peripheral vaso constriction which occurs as a sequel to body's response to pyrogen which leads to two mechanisms in the body i.e. increased heat production and decreased heat loss from the body. When there is peripheral vaso dilatation, the person may have profuse sweating, the person likely to have aversion to the hot atmosphere. Thus, when the individual is having peripheral vasoconstriction, he will have frequent liking towards hot substances and when he is having peripheral dilatation, he will have liking towards cold substances.
 As far as Ayurveda fundamentals are concerned, there will be increase of Shita quality in the body, thus, it indirectly responsible for the liking of Ushna quality substances. When there is decrease of Shita quality in the body, then opposite mechanism happens i.e. aversion to the Ushna substances.
- 32 This premonitory symptom is seen in the patient of Vataja Jvara. Yawning is an activity consists of wide opening of mouth with maximal widening of jaw, together with a long and deep inhalation through mouth and nose, followed by slow expiration, associated with a feeling of comfort. The average duration of yawn is 5 seconds. Additionally, stretching of the limbs also frequently accompanies yawning in humans. Following hypotheses can be put forth in understanding the concept of yawning.
- When an individual does yawning, that will lead to the shift of temperature from the core of the head region to the periphery of head and neck.
 - Yawning relieves the pain in the ears by opening of eustachian tube which has a connection with middle ear by removing the blockage present in the tube.
 - Yawning is triggered when blood or brain oxygenation is insufficient, when O2 level is decreased and CO2 is increased. Yawning removes CO2 from the lungs and increases the O2 circulation in the brain.
- 33 This symptom is attributable myalgia or shivering which manifests in an individual who has early manifestation of fever.
- 34 This is attributable to the dominance of Ama in the body. This can be either related to alimentary canal or any part of the body. In this occasion, the macromolecules or intermediate molecules will move slowly in the alimentary canal, thus the person may have heaviness of the abdomen.
 When there is pyrogen in any part of the body, that will be responsible for slower movement of the blood in the capillaries of the area of affliction, this is termed as sludging of blood. This is responsible for the heaviness of the body parts.
 Sympathetic stimulation causes the arrector pili muscles attached to the hair follicles to contract, which brings the hairs to upright stance. This mechanism occurs when the body is too cold. This occurs along with skin vaso constriction and heat production in the body.
- 35 This is also termed as piloerection i.e. hairs standing on end. Sympathetic stimulation causes the arrector pili muscles attached to the hair follicles to contract, which brings the hairs to upright stance. This mechanism occurs when the body is too cold. This occurs along with skin vaso constriction and heat production in the body.

Ca.I. – 1/10

- *Aruchi*³⁶: tastelessness
- *Tama*³⁷: darkness in front of the eyes
- *Apraharsha*: dissatisfaction
- *Shita*³⁸: feeling of coldness
- *Alasya*³⁹: laziness
- *Sasra Nayana*⁴⁰: reddish discolouration of the eyes
- *Klama*⁴¹: fatigue without doing activities
- *Jvalana*⁴²: burning sensation
- *Vayu Ambu Anischita Bhakti Dvesha*⁴³: affinity and aversion towards exposure to air and consumption of water infrequently
- *Avipaka*⁴⁴: indigestion
- *Balahani*⁴⁵: reduced⁴⁶ or loss of strength
- *Varna*⁴⁷ hani: loss of complexion
- *Shila Vikriti*⁴⁸: changes in behaviour

Bheda of Jvara:

2 types: Sharirika and Manasika

- 36 This symptom is attributable to sympathetic nervous system stimulation which in turn leads to reduced secretions of alimentary canal including oral cavity.
- 37 This symptom is attributable to hypovolemia occurs in a patient of fever.
- 38 This is attributable to peripheral vaso constriction.
- 39 This is attributable to the presence of Ama in the body. When an individual is having the entry of pathogen in the body, there will be slow circulation in the body which is called sludging of blood. When there is poor circulation in any part of the body, the person will have heaviness of the body and this may eventually lead to laziness of the individual.
- 40 This is attributable to allergic exposure to the eyes and thus causing inflammatory reaction in the eyes which is responsible for local rise of temperature. This indicates the dominance of Pitta in the body.
- 41 This symptom is attributable to poor blood circulation to the skeletal muscles owing to peripheral vaso constriction.
- 42 This is attributable to the increase of Pitta Dosha in the body. This may be probably due to the increased blood circulation to the periphery and this occurs when an individual is having increased temperature inside the body. This may lead to excess perspiration which may increase from normal 400 ml/ 24 hours to 2 litres/ hour. This may lead to the appearance of burning sensation of the body.
- 43 This may be attributable to infrequent demand and supply of water to the body. This may indirectly suggest inadequate supply of water owing to infrequent loss of water from the body.
- 44 This is attributable to the sympathetic nervous system activation and this is leading to the poor secretion in the alimentary canal and thus, the person may have indigestion.
- 45 This is attributable to poor metabolic activity in the body. Because of poor metabolism, the person may have poor absorption of glucose, water and electrolytes such as sodium, potassium, calcium ions which are the major causes of reduced strength in the body.
- 46 This is attributable to poor metabolic activity in the body. Because of poor metabolism, the person may have poor absorption of glucose, water and electrolytes such as sodium, potassium, calcium ions which are the major causes of reduced strength in the body.
- 47 वर्णभेदेन ग्लानिहर्षरीक्ष्यस्नेहा व्याख्याताः ॥१०॥
वर्णभेदेनेति यथा वर्णविभागेन रिष्टं, तथा ग्लान्यादिनिष्ठा गेनापीत्यर्थः।
इपे इहोपचयो हेयः, मानसहर्षस्नेह चानुपा चिकारेऽसङ्गतत्वात् ॥१०॥
वर्णशब्देन च वर्णसहचरिता- बक्षुर्भाया रीक्ष्यादयोऽपि गृह्यन्ते।
The term Varna refers to the abnormal texture of the skin such as dryness which is perceivable by eyes. Thus, if the person is suffering from early manifestation of fever, there will be dryness of the skin due to the poor peripheral circulation of the blood.
- 48 Whenever the internal body temperature becomes too high, signals from the temperature controlling areas in the brain give the person a psychic sensation of being overheated. Conversely, whenever the body becomes too cold, signals from the skin and probably also from deep receptors elicit the feeling of cold discomfort. Therefore, the person makes appropriate environmental adjustments to reestablish the comfort, such as moving into a heated room or wearing well insulated clothing in freezing weather.
- 49 This type of classification of Jvara is made based on the dominance of Shita and Ushna Virya in the body.
- As per the current physiological understanding, raised body temperature is possible because of two pathological mechanisms – increased heat production or reduced heat loss from the body.
 - Whenever, there is increased heat generation from the body, it may be associated with peripheral vasoconstriction and muscle twitchings which is identified as chills or rigors. This is attributable to Ushnabhiprayaja Jvara.
 - When there is hampered heat loss from the body, it implies presence of more heat inside the body. In this occasion, the person desirous of having more frequent or continuous contact with cold substances. This is attributable to Shitabhiprayaja Jvara.

शारीरो जायते पूर्व देहे, मनसि मानसः।

- In *Sharirika Jvara*, first there will be physical symptoms, followed by mental dysfunction. Ca.Ci. – 3/36

ज्वरप्रत्यात्मिकं लिङ्गं संतापो देहमानसः।

In *Sharirika Jvara*, the person will have raised body temperature. This is observable by the physician. This is followed by dysfunction of psyche. Ca.Ci. – 3/31

- In case of *Manasika Jvara*, first there will be mental symptoms, followed by physical dysfunction. Ca.Ci. – 3/36

वैचित्त्यमरतिग्लानिर्मनसस्तापलक्षणम्॥

In *Manasika Jvara*, the person will be presented with abnormal functions of *Chitta*, restlessness and mental exhaustion. These *Lakshana* are followed by physical symptoms. This is typical of *Manasika Jvara*.

2 types: *Saumya* and *Agneya Jvara*⁴⁹

वातपित्तात्मकः शीतमुष्णं वातकफात्मकः ॥३७॥

इच्छत्युभयमेतत्तु ज्वरो व्यामिश्रलक्षणः।

- In case of *Vatakapahaja Jvara*, the person will be having Ca.Ci. – 3/37-38

Ca.I. - /10, along with Chakrapani commentary

Chakra, Ca.I.: 1/3

the symptoms related to increase of *Shita Guna* in the body. In such situation, the person will be presented with liking towards consumption of hot substances. That is why, this type of *Jvara* is called *Saumya Jvara* or *Ushnabhiprayaja Jvara*.

- In case of *Vatapittaja Jvara*, the person will be having the symptoms related to increase of *Ushna Guna* in the body. In such situation, the person will be having liking towards consumption of cold substances. That is why, this type of *Jvara* is called *Agneya Jvara* or *Shitabhiprayaja Jvara*.

2 types: *Antarvega* and *Bahirvega Jvara*

अन्तर्दाहोऽधिकस्तृष्णा प्रलापः श्वसनं भ्रमः॥३९॥

सन्ध्यस्थिशूलमस्वेदो दोषवचविनिग्रहः॥

अन्तर्वेगस्य लिङ्गानि ज्वरस्यैतानि लक्षयेत्॥४०॥

संतापोऽध्यधिको बाह्यस्तृष्णादीनां च मार्दवम्॥

बहिर्वेगस्य लिङ्गानि सुखसाध्यत्वमेव च॥४१॥

Ca.Ci. – 3/39-41

- ***Antarvega Jvara*⁵⁰**: In case of *Antarvega Jvara*, the person will be presented with burning sensation, excessive thirst, irrelevant speech, breathlessness, giddiness, pricking type of pain in the joints and bones, absence of sweating, absence of movement of vitiated *Dosha* from the body, constipation. This type of *Jvara* is difficult to cure. This is because the *Lakshana* are not easily observable by the physician.

- ***Bahirvega Jvara*⁵¹**: In *Bahirvega Jvara*, the person will be presented with raised body temperature in the periphery, the symptoms such as thirst etc will become milder. This type of *Jvara* is considered to be easily curable as the *Lakshana* are clearly identifiable.

2 types: *Prakrita* and *Vaikrita Jvara*:

प्राकृतः सुखसाध्यस्तु वसन्तशरदुद्भवः॥

Ca.Ci. – 3/42

कालप्रकृतिमुद्दिश्य निर्दिष्टः प्राकृतो ज्वरः॥४४॥

प्रायेणानिलजो दुःखः कालेष्वन्येषु वैकृतः॥

Ca.Ci. – 3/48-49

- ***Prakrita Jvara*⁵²**: The term *Prakrita Jvara* refers to the aggravation of symptoms of *Jvara* in specific time and specific *Prakriti* individual. For example, aggravation of fever indicating the dominance of *Kapha Dosha* in the body in *Kapha Prakriti* individual in the morning

time or in *Vasanta Rtu* is *Prakrita Kaphaja Jvara*. *Pitta* dominant *Jvara* occurring *Pitta Prakriti* individual in *Sharat Rtu* is considered as *Prakrita Pittaja Jvara*. *Vataja Jvara* in *Vata Prakriti* individual in *Varsha Rtu* is considered as *Prakrita Vataja Jvara*.

- ***Vaikrita Jvara*⁵³**: Manifestation of *Kaphaja Jvara* in the season other than *Vasanta Rtu* is termed as *Vaikrita Kaphaja Jvara*. Manifestation of *Pittaja Jvara* in the season other than *Sharat Rtu* is termed as *Vaikrita Pittaja Jvara*. Manifestation of *Vataja Jvara* in the season other than *Varsha Rtu* is termed as *Vaikrita Vataja Jvara*.

2 types: *Sadhya* and *Asadhya Jvara*:

बलवत्स्वल्पदोषेषु ज्वरः साध्योऽनुपद्रवः॥

Ca.Ci. – 3/49

- ***Sadhya Jvara***: when the person is having following situations, then such *Jvara* is considered to be *Sadhya* (curable). This indicates the mild⁵⁴ vitiation of *Dosha*, duration of exposure of etiological factors for a shorter time period and involvement of *Dhatu* in disease process is minimum.

- If the *Rogibala* is very strong than *Rogabala*
- If the vitiation of *Dosha* in the individual is mild in nature
- If the *Jvara* is not associated with any *Upadrava* (complications)

- ***Asadhya Jvara***: This is considered to be incurable type of *Jvara*. The following are the situations, where this type of *Jvara* is observed.

हेतुभिर्बहुभिर्जातो बलिभिर्बहुलक्षणः॥५०॥

ज्वरः प्राणान्तकृद्यश्च शीघ्रमिन्द्रियनाशनः॥

Ca.Ci. – 3/50

सप्ताहाद्वा दशाहाद्वा द्वादशाहात्तथैव च॥५१॥

सप्रलापभ्रमश्वासस्तीक्ष्णो हन्याड्वरो नरम्॥

Ca.Ci. – 3/51

ज्वरः क्षीणस्य शूनस्य गम्भीरो घेरात्रिकः॥५२॥

असाध्यो बलवान् यश्च केशसीमन्तकृज्वरः॥

Ca.Ci. – 3/52

- If the manifestation of *Jvara* is occurring due to exposure to multiple aetiological factors. For example, when an individual is exposed to two etiological factors such as *Prapaka* (suppuration) stage and *Ajirna* (indigestion) stage, we can observe the dominance of *Pitta* and *Kapha Dosha* at a particular time, that means

- 50 In *Antarvega Jvara*, there will be peripheral vaso constriction. Because of this, the person will be having cold sensation from the exterior and core temperature is maintained or raised. As there is peripheral vaso constriction, the person will have absence of sweating. As there is sympathetic nervous stimulation, the person will have thirst, irrelevant speech and constipation.
- 51 In *Bahirvega Jvara*, there will be peripheral vaso dilatation. That is why the person will have excess amount of sweating which leads to the transfer of excess amount of heat from the body to the exterior. Due to the relaxation of sympathetic nervous system, the person will have mild appearance of thirst and mild appearance of other *Lakshana* of *Antarvega Jvara*.
- 52 *Prakrita Jvara* manifested due to the vitiation of *Kapha* and *Pitta Dosha* is easily curable. *Prakrita Vataja Jvara* is difficult to cure. The reason is in case of *Jvara*, the line of treatment is *Langhana* or *Apatarpana* (measures which will help in the lightness of the body). This measure will lead to the vitiation of *Vata Dosha*, but pacify vitiated *Kapha* or *Pitta Dosha*.
- 53 *Vaikrita Jvara* is invariably difficult to cure as the nature of vitiated *Dosha*, *Prakriti* of the individual and season in which symptoms of *Jvara* are not seen in association with one another.
- 54 Let us take the example of two patients suffering from *Kaphaja Jvara*. First person is having the history of consumption of curds of 100 ml since 3 days and second person is having the history of consumption of curds of 200 ml for 4 days and grapes of 50 gms since 2 days. Here, the first person is having mild vitiation of *Kapha Dosha* and second person is having moderate vitiation of *Kapha Dosha*.

the physician has to manage the vitiations of two *Dosha*, the management of these *Dosha* are entirely different. In that situation, management of the disease in that individual will be difficult.

• If *Rogabala* (virulence of disease) of the individual is more potent than *Rogibala* (physical strength of the individual). For example, *Svedavarodha* (absence of sweating) in one part of the body and duration of the presenting complaint is more than 5 days. In another patient presented with *Svedavarodha* (absence of sweating) all over the body and the duration of the symptoms is more than 7 days. As far as virulence of the disease is concerned, in the former individual, the person is having mild *Rogabala* and in the latter individual, the person is having more *Rogabala* than the strength of the person himself. In the latter case, the manifestation of *Jvara* is incurable when compared to former individual.

• If the person is presented with multiple *Lakshana* (symptoms). For example, if the person is having all *Lakshana* mentioned in *Jvara*, *Svedavarodha* (absence of sweating), *Santapa* (raised body temperature) and *Sarvanga Grahana* (stiffness of body parts), then such disease is considered to be incurable in nature.

• If the person is presented with life threatening symptoms, for example if the person is presented with rapid manifestation of *Trishna* (thirst) or *Moha* (confusion) indicate progression of the illness into rapid *Udaka Kshaya* (*hypovolemia*⁵⁵). If proper management of this state is not being taken, then it will lead to poor prognosis.

• If the person is presented with rapid deterioration of functions of *Indriya*⁵⁶. In this situation, there will be redistribution of the fluid from the periphery to the core of the body and this is secondary to hypovolemia.

• If the person is suffering from fever from 7 days in *Vataja Jvara*, 10 days in *Pittaja Jvara* and 12 days in *Kaphaja Jvara* and associated with irrelevant speech in *Vataja Jvara*, giddiness in *Pittaja Jvara* and breathlessness in *Kaphaja Jvara*.

• If the person is presented with *Dhatu Kshaya* (depletion of body tissues). In this condition, the person will have hypovolemia.

• If the person is having the swelling (localised or generalised). In case of generalised swelling in the individual, the person will have reduced fluid compartment in intravascular compartment and more fluid accumulation in the interstitial compartment. This implies *hypovolemia*⁵⁷.

• If there is involvement of deeper body tissues (*Dhatu*). This will lead to more depletion of body tissues and this is leading to hypovolemia.

• If the person is presented with symptoms which is observed from *longer duration*⁵⁸ or persist for longer duration in night time

• If the person is presented with matted *hair*⁵⁹ appearance

Ama, Pachyamana and Nirama Jvara Lakshana

Ama Jvara:

अरुचिश्चाविपाकश्च गुरुत्वमुदरस्य च॥1 3 3॥

हृदयस्थाविशुद्धिश्च तन्त्रा चालस्यमेव च।

ज्वरोऽविसर्गी बलवान् दोषाणामप्रवर्तनम्॥1 3 4॥

लालाप्रसेको हल्लासः क्षुनाशो विरंसं मुखम्।

स्तब्धसुप्तगुरुत्वं च गात्राणां बहुमूत्रता॥1 3 5॥

न विड् जीर्णां न च ग्लानिर्ज्वरस्यामस्य लक्षणम्। Ca.Ci. - 3/133-136

ज्वरोपद्रवतीक्षणत्वमग्लानिर्बहुमूत्रता।

न प्रवृत्तिर्न विड् जीर्णां न क्षुत्सामज्वराकृतिः॥5 4॥ A.Hr.Ni. - 2/54

55 Normally, 400 ml of sweat is formed in 24 hours, but when an individual is having high degree of fever, there will be loss of fluid of about 2 litres/ hour. This mechanism is responsible for rapid development of thirst and confusion secondary to reduced hypoperfusion of brain. In this situation, along with water, even sodium ions will be lost. It indicates rapid replenishment of fluid and sodium ions is necessary.

56 Let us take the example of *Rasanendriya* (tongue). When the person is suffering from fever, first of all there will be loss of fluid and electrolytes, sodium in particular. In this situation, the tongue will receive minimum blood supply to maintain the blood supply to the heart and brain which are the two main organs need continuous blood supply. Because of the poor salivary secretion in the oral cavity, the person may have poor taste perception. In the same way, one has to think for *Chakshurindriya*, *Sparshanendriya*, *Ghranendriya*, *Shrotrendriya* also. The same can be applied to the *Karmendriya* (*Vak*, *Pada*, *Pani*, *Payu* and *Upastha*) as well.

57 In this situation, the person will have reduced protein content in the intravascular compartment and increased protein content in the interstitial compartment.

58 Whenever an individual is suffering from fever for longer duration, the following points can be deduced:

- Increased basal rate of metabolism of all cells of the body
- Extra rate of metabolism caused by muscular activity including muscle contractions caused by shivering
- Extra metabolism caused by the effect of thyroxine on the cells
- Extra metabolism caused by the effect of epinephrine, norepinephrine and sympathetic stimulation on cells
- Extra metabolism caused by increased chemical activity in the cells themselves, especially when the cell temperature increases
- Extra metabolism needed for digestion, absorption and storage of food
- Slowness in the conduction of heat to the periphery from the core of the body
- Slowness in the transfer of heat from the skin to the exterior

Because of above said reasons, the person may have emaciation of the body.

59 When an individual develops matted hair appearance, it indicates increased perspiration in the head too. This is responsible for the dryness of the scalp region and in turn responsible for matted appearance of hair.

The term *Ama Jvara*⁶⁰ refers to the state of *Jvara* where vitiation of *Dosha* observed in *Amashaya* of *Mahasrotas*. This is observed in the initial phase of *Jvara* manifestation.

- ✱ **Aruchi**⁶¹ – tastelessness
- ✱ **Avipaka**⁶² – improper digestion
- ✱ **Udarasya Gurutva**⁶³ – heaviness of the abdominal region
- ✱ **Hridayasya Avishuddhi**⁶⁴ – discomfort in the chest
- ✱ **Tandra** – drowsiness
- ✱ **Alasya** – laziness
- ✱ **Jvara which is Avisargi**⁶⁵ – continuous fever
- ✱ **Balavan** – Rogabala is more
- ✱ **Doshanam Apravartana**⁶⁶ – no expulsion of vitiated *Dosha* from the body such as absence of vomiting or diarrhoea or expectoration etc.

- ✱ **Lala Praseka**⁶⁷ – excess salivation
- ✱ **Hrillasa**⁶⁸ – nausea
- ✱ **Kshudha Nasha** – loss of appetite⁶⁹
- ✱ **Virasa Mukha** – abnormal taste perception
- ✱ **Stabdhata, Suptata and Gurutva of Gatra** – stiffness, numbness and heaviness of the body parts
- ✱ **Bahumutrata** – excess formation of urine⁷⁰
- ✱ **Na Vit** – no formation of faecal matter or constipation⁷¹
- ✱ **Na Jirna** – absence of digestion⁷²
- ✱ **Glani**⁷³ – exhaustion

Pachyamana Jvara:

ज्वरवेगोऽधिकस्तृष्णा प्रलापः श्वसनं भ्रमः॥१३६॥

मलप्रवृत्तिरुत्क्लेशः पच्यमानस्य लक्षणम्।

Ca.Ci. – 3/136-137

- 60 The Lakshana such as Aruchi, Avipaka, Udarasya Gurutva, Hridayasya Avishuddhi, Tandra and Alasya can be attributable to the presence of Ama in the Mahasrotas, that too Amashaya in particular.
- 61 This can be attributable to the over stimulation of parasympathetic nervous system. This is responsible for the over secretion of salivary secretion. When there is activation of sympathetic nervous system, the person may have tastelessness because of poor salivary secretion which is required for tastebuds to function.
- 62 Avipaka or indigestion can be attributable to either poor gastric secretion or because of poor peristaltic movements or because of poor gastric emptying.
- 63 This is attributable to the excess presence of macromolecules in the stomach, duodenum, small intestine or large intestine. This occurs as a result of decreased digestive enzymes secretion in the alimentary canal.
- 64 This is attributable to the presence of macromolecules in the stomach and poor peristaltic movements leading to the delay in emptying of the stomach.
- 65 When there is continuous fever in an individual, it indicates continuation of pathological events of fever inside the body i.e. either increased heat production or hampered heat loss from the body.
- 66 When there is over stimulation of parasympathetic nervous system which is associated with inhibitory signals to the sympathetic nerve fibres supplying the stomach and signals from the duodenum are resulting in delaying in the emptying of pyloric antrum. This is further responsible for the poor peristaltic movements in the alimentary canal.
- 67 This is attributable to the over activation of parasympathetic nervous system. When there is activation of parasympathetic nervous system on salivary glands, there will be excess salivation. In the stomach and small intestine, there will be excess mucus secretion thus, the person may have loss of appetite.
- 68 The presence of this complaint indicates the distension of either stomach or duodenum. This may occur as a result of delayed pyloric emptying or poor peristaltic movements of the stomach secondary to poor conversion of macromolecules to intermediate to micromolecules.
- 69 This is attributable to increase in secretion which in turn responsible for the deactivation of pepsinogen which occurs as a result of poor hydrochloric acid secretion. In this situation, if an individual consumes food, he will have poor digestion because of poor activation of pepsinogen.
- 70 When the person is having Ama Jvara, following mechanisms occur in the body.
 1. Poor blood circulation in the alimentary canal
 2. Poor peripheral circulation secondary to peripheral vasoconstriction
 3. There will be raised body temperature in the body secondary to either increased heat production in the body or hampered heat loss from the body
 4. As the time passes, the above mechanisms will lead to heat loss either urine and breath. Thus, the person may develop excess micturition from the body.
- 71 This is attributable to the stimulation of sympathetic nervous system. When there is activation of sympathetic nervous system, closure of sphincters in the alimentary canal i.e. upper oesophageal sphincter, lower oesophageal sphincter, pyloric sphincter, ileocaecal junction and anal sphincters occur. Thus constipation is one of the symptoms observed in the patient who has the history of recent fever.
- 72 This is attributable to the stimulation of sympathetic nervous system. When there is activation of sympathetic nervous system, along with closure of sphincters, reduced secretion in the alimentary canal occur. This will lead to the presenting complaint of indigestion. This is the status of Agnimandya and Ajirna in the person suffering from Jvara.
- 73 When an individual is presented with exhaustion, it implies that the individual is having poor digestive secretion. The above said presenting complaint can occur because of
 1. Poor digestive function owing to poor carbohydrate absorption
 2. Increased formation of metabolic acidosis (ketoacidosis and lactic acidosis) secondary to poor absorption
 3. This may be associated with increased muscular metabolism (chills or rigor)
 Because of above said reasons, person may have inadequate nutrients to the each and every cell of the body. Thus, the person may have exhaustion

ज्वरवेगोऽधिकं तृष्णा प्रलापः श्वसनं भ्रमः।

मलप्रवृत्तिरुत्क्लेशः पच्यमानस्य लक्षणम्॥55॥ A.Hr.Ni. - 2/55

In case of *Pachyamana Jvara*, the *Samprapti* of the disease is little bit prolonged and the *Lakshana* of the disease are comparatively severe than *Ama Jvara*. All the *Lakshana* mentioned in the following may not be visible in suffering individual, depending upon the dominance of *Dosha* responsible for the manifestation of the symptoms, fewer symptoms can be manifested. These are manifested after the appearance of *Ama Jvara*, can be observed before the appearance of *Nirama Jvara*.

- **Adhika Trishna**⁷⁴ – excessive thirst
- **Pralapa**⁷⁵ – irrelevant speech
- **Shvasana**⁷⁶ – breathlessness
- **Bhrama**⁷⁷ – giddiness
- **Mala Pravritti**⁷⁸ – passage of stools
- **Utklesha**⁷⁹ – nausea or vomiting

Nirama Jvara:

भुत् क्षामता लघुत्वं च गात्राणां ज्वरमार्दवम्॥137॥

दोषप्रवृत्तिरष्टाहो निरामज्वरलक्षणम्।

जीर्णतोमविपर्यासात्सप्तरात्रं च लङ्घनात्।

Ca.Ci. - 3/137-138

A.Hr.Ni. - 2/56

The term *Nirama Jvara* indicates the stage of *Jvara* where there is decrease of *Amatva* in the body. This is indicative

of restoration of the disease process in the body.

- **Kshut Kshamata** – appetite⁸⁰
- **Laghutva**⁸¹ – lightness of the body
- **Gatra Mardava**⁸² – softness of the body parts
- **Jvara Mardava** – symptoms indicating the recovery from the fever
- **Dosha Pravritti**⁸³ – expulsion of vitiated *Dosha* from the body i.e. vomiting, diarrhoea or expectoration etc
- **Ashta Aha** – 8th day of appearance of fever. This term refers to even in the absence of *Kshut Kshamata* (appearance of appetite), when the duration of *Jvara* crosses 8th day in appearance, we have to consider such *Jvara* as *Nirama Jvara*.

Vatabalasaka Jvara: The term *Vatabalasaka* is comprised of two words. They are *Vata* (*Vata* dominant) and *Balasa* (*Kapha* dominant). The clinical features of *Vatabalasaka Jvara* are

नियं मन्दज्वरो रुक्षः शुमकस्तेन सीदति।

स्तप्याङ्गः श्रेष्मभूयिष्ठो नरो वातपडालकी॥40॥

Ma.Ni. - 2/40

- **Nitya Manda Jvara**⁸⁴ – slight raise of body temperature
- **Ruksha**⁸⁵ – increased dryness of the body or body parts or skin
- **Shunaka**⁸⁶ – swelling either localised or generalised
- **Stabdhanga**⁸⁷ – stiffness of the body or body parts

This occurs as a result of hypovolemia develops. When we see the textbooks of physiology, normally, 400 ml of sweat is formed in 24 hours, but when an individual is having fever, the amount of sweat formed may increase upto 2 litres/ hour. This indicates the rapid development of dehydration, which exhibits as a symptom of thirst.

- 74 This is attributable to further progression of pathological events in the diseased individual. This occurs as a result of hypovolemia develops. When we see the textbooks of physiology, normally, 400 ml of sweat is formed in 24 hours, but when an individual is having fever, the amount of sweat formed may increase upto 2 litres/ hour. This indicates the rapid development of dehydration, which exhibits as a symptom of thirst.
- 75 This is attributable to dehydration which may be responsible for either poor cerebral circulation or electrolyte imbalances, hyponatremia in particular. Along with loss of 2 litres of water per hour, there will be loss of sodium ions also. It is being said that in an unacclimatised person, the person loses 15 to 30 gms of sodium salts each day for first few days.
- 76 This can be attributable to the person is having heat loss from the expired air. Whenever, the individual is having poor heat loss at the beginning, this indicates peripheral vasoconstriction, shivering and increased metabolism from different parts of the body or there will be action of WBCs on pathogens. When the thermoregulatory setpoint is reached, this may be responsible for either increased sweat production or increased expiration of warmer or hot air or excess sweating.
- 77 This is attributable to hypovolemia leading to poor circulation of blood to head region. This may occur as a result of hypoglycemia or hyponatremia as well.
- 78 This may be attributable to excess secretion of mucus from the large intestine, which contains bicarbonate ions in abundance owing to the activation of parasympathetic nervous system.
- 79 This may be attributable to excess secretion in the stomach which leads to stretching of stomach, thereby the person may have pyloric antrum closure leading to reverse peristaltic movements, the end result may be either nausea or vomiting.
- 80 This is attributable to the activation of parasympathetic nervous system. Activation of parasympathetic nervous system is responsible for the relaxation of sphincters in the alimentary canal and secretions in the alimentary canal.
- 81 When there is subsidence of fever, replenishment of body fluids starts to occur, this will lead to the disappearance of mechanism of sludging of blood. Thus, the person will have lightness of the body.
- 82 This can be attributable to replenishment of body fluids. When the person is having excess fluid loss, which is an essential feature of fever, there will be slowness of movement of blood in the smaller vessels, which is termed as sludging of blood. This is responsible for the appearance of heaviness of the body parts.
- 83 This is attributable to relaxation of sphincters of different areas of gastrointestinal tract which is presented with vomiting or diarrhoea etc.
- 84 Slight raise in body temperature indicates early onset of pathogenic events.
- 85 The presence of dryness of body or body parts, indicates the peripheral vaso constriction, which imply poor blood circulation in the peripheral blood vessels.
- 86 The term *Shunaka* refers to either localised or generalised swelling. This kind of presenting complaint may imply that increased protein content in the interstitial and intracellular compartments. By applying the Sterling forces, we can infer that excess concentration of proteins in interstitial and intracellular compartments, more fluid will move from capillaries to interstitial compartments.
- 87 The term *Stabdhanga* refers to either localised or generalised stiffness. This kind of presenting complaint may imply that increased protein content in the interstitial and intracellular compartments, more fluid will move from capillaries to interstitial compartments.

- **Kapha bhuyishta⁸⁸** – predominance of Kapha Dosha seen in the patient

pralepaka Jvara: Cardinal features of *Pralepaka Jvara* are as follow:

प्रतिष्पिक्ष गात्राणि धर्मेण गौरवेण च।

मन्द्रङ्गवरविलेपी व सशीतः स्यात्प्रलेपकः॥

Ma.Ni. – 2/41

- This is a Kapha Pitta dominant condition
- Sensation of coating on the whole body or body parts⁸⁹
- Heaviness of the body or body parts⁹⁰
- **Mandajvara⁹¹** – slight raise in body temperature
- **Sheeta⁹²** – sensation of coldness of the body or body parts

Agantuja Jvara:

आगन्तुरष्टमो यस्तु स निर्दिष्टश्चतुर्विधः॥११॥

अभिघाताभिषङ्गाभ्यामभिचाराभिशापतः।

Ca.Ci. – 3/111-112

Agantuja Jvara is manifested due to the external causes. In this situation, first there will be appearance of symptoms which will be followed by the vitiation of *Doshas*. The time period between the appearance of symptoms and vitiation of *Dosha* in *Agantuja Jvara* is 7 days. Four types of *Agantuja Jvara* are explained. They are

1. *Abhighataja Jvara*
2. *Abhishangaja Jvara*
3. *Abhicharaja Jvara*
4. *Abhishapaja Jvara*

Abhighataja Jvara⁹³:

शस्त्रलोष्टकशाकाष्ठमुष्टय नितलद्विजैः॥१२॥

तद्विधैश्च हते गात्रे ज्वरः स्यादभिघातजः।

तत्राभिघातजे वायुः प्रायो रक्तं प्रदूषयन्॥१३॥

सव्यथाशोफ वैवर्ण्यं करोति सरुजं ज्वरम्।

Ca.Ci. – 3/112-114

- may complain of chills or rigors. This is usually indicative of heat production which occurs as a result of muscular twitches. It indicates that the temperature produced during the pathogenesis, is inadequate to combat the pyrogen.
- 88 This is commonly seen in the Anupa Desha. As per Ayurveda fundamentals, this is identified by ample amount of water in the environment, presence of greenery around the environment. This may be the breeding place of mosquitoes. This may be the place of mosquito borne infections or water or contaminated water borne diseases.
- 89 No possible explanation can be given to this symptom from the author.
- 90 Heaviness of the body parts imply poor movement of the blood. This can be considered to be due to sludging of blood. When any pathogen enters into the body, that will be responsible for the increase of inflammatory markers in the body which are nothing but proteins. This is responsible for the increase of oncotic pressure in the interstitial compartment. As the movement of fluid, WBC etc move into the interstitial compartment. This will lead to the slow movement of the blood from the venous end of capillaries. Thus, the person may have heaviness of the body parts.
- 91 This may imply initial phase of pyrogenesis or it may imply low virulence of the disease pathogen.
- 92 This may imply there will be peripheral vaso constriction. This is responsible for the coldness of the body parts. In case of cold air exposure, the person may have dryness of respiratory passages because of respiratory passages.
- 93 In this situation, the person will have physical trauma which leads to the discontinuity of the skin or epithelial membrane. This may lead to the entry of the microorganisms into the injured tissue if no proper measures are taken to prevent the haemorrhage or to prevent contamination.
- 94 This may be attributable to exposure of the body to toxic plants or toxic gases or allergic plants. This may also be attributable to extreme emotional disturbances which may be leading to autonomic dysfunction, sympathetic nervous system in particular.

- *Abhighataja Jvara* manifests due to the injury to the body because of trauma to the body by weapons, small stones or fist etc

- In this type of *Jvara*, first there will be vitiation of *Vata Dosha* along with diminution of *Rakta* from the body.

- This is associated with *Vyatha* (discomfort), *Shopha* (swelling), *Vaivarnya* (discolouration) and *Saruja* (pain)

Abhishangaja Jvara⁹⁴:

कामशोकभयक्रोधैरभिषक्तस्य यो ज्वरः॥१४॥

सोऽभिषङ्गाज्वरो ज्ञेयो यश्च भूताभिषङ्गजः।

कामशोकभयाद्वायुः क्रोधात् पित्तं, त्रयो मलाः॥१५॥

भूताभिषङ्गात् कुप्यन्ति भूतसामान्यलक्षणाः।

भूताधिकारे व्याख्यातं तदष्टविधलक्षणम्॥१६॥

विषवृक्षानिलस्यशान्तिथाऽन्यैर्विषसंभवेः।

अभिषक्तस्य चाप्याहुर्ज्वरमेकेऽभिषङ्गजम्॥१७॥ Ca.Ci. – 3/114-117

- This type of *Jvara* occurs due to the exposure of the individual to poisonous plant, impure air, toxic substances, mental factors such as lust, grief, fear and anger; exposure to evil spirits.

- Exposure to lust, grief and fear will lead to the vitiation of *Vata Dosha*; exposure to anger will lead to the vitiation of *Pitta Dosha* and exposure to evil spirits will lead to the vitiation of all three *Dosha*.

ध्याननिःश्वासबहुलं लिङ्गं कामज्वरे स्मृतम्॥१८॥

शोकजे वाप्यबहुलं त्रासप्रायं भयज्वरे।

क्रोधजे बहुसंरम्भं भूतावेशे त्वमानुषम्॥१९॥

मूर्च्छामोहमदग्लानिभूयिष्ठं विषसंभवे।

केषाञ्चिदेषां लिङ्गानां संतापो जायते पुरः॥२०॥

Ca.Ci. – 3/122 – 124

- In case of *Kamaja Jvara*, the person will be presented with *Dhyana* (absent mindedness or thinking continuously towards one object) and *Nishvasa*

*Bahula*⁹⁵ (excess expiration or pursed lip breathing)

- In case of *Shokaja Jvara*, the person will be presented with *Bashpa Bahula* (increased tears from the eyes)
- In case of *Bhayaja Jvara*, the person will be presented with *Trasa Praya* (fearful look)
- In case of *Krodhaja Jvara*, the person will be presented with angry look
- In case of *Bhutaveshaja Jvara*, the person will be having inhuman activities.
- In case of *Vishaja Jvara*, the person will be having *Murcha* (altered state of consciousness); *Moha* (confusion); *Mada* (intoxication), *Glani* (fatigue or exhaustion)

Abhicharaja and Abhishapaja Jvara:

चित्तेन्द्रियशरीराणामर्तयोऽन्याश्च नैकशः।

प्रयोगं त्वभिचारस्य दृष्ट्वा शापस्य चैव हि॥120॥

स्वयं श्रुत्वाऽनुमानेन लक्ष्यैते प्रशमेन वा।

वैविध्यादभिचारस्य शापस्य च तदात्मके॥121॥

Ca.Ci. – 3/120 – 121

- *Abhicharaja Jvara* is a type of *Jvara* manifests in an individual due to performing rituals mentioned in *Atharvaveda* etc
- *Abhishapaja Jvara* is the type of *Jvara* manifests in an individual due to the curse of the heavenly bodies, preceptors and elderly people.
- Above two types of *Agantuja Jvara* are considered to be manifested in the individuals due to the dominance of all three *Dosha*.

Vishama Jvara:

विषमो विषमारम्भक्रियाकालोऽनुपङ्गवान्।

A.Hr.Ni. – 2/69

अधुना विषमज्वरस्य स्वरूपं प्रतिपादयन्नाह – विषम इति।

विषमो – विषमसंज्ञो ज्वरो, विषमारम्भक्रियाकालो भवति।

आरम्भश्च क्रिया च कालश्च आरम्भक्रियाकालः।

विषमा आरम्भक्रियाकाला यस्य स एवम्।

तत्र विषम आरम्भो यथा, – कश्चित्पूर्वं मूर्धानमारभ्यागच्छति, कश्चित्पृष्ठम्, कश्चि- जङ्घाभ्यामिति।

विषमक्रियो यथा, – कश्चिच्छीतकृत् कश्चिद्वाहकृदिति।

विषमकालश्च यथा, – कश्चित्पूर्वाह्णे समागच्छति, काश्च न्मध्याहे, कश्चिदपराह्णे, कश्चिन्निशीथे, इति।

तथा, अनुपङ्गवान्दीर्घकालानुबन्धी। Arunadatta, A.Hr.Ni. – 2/69

*Vishama Jvara*⁹⁶ is characterised by irregularity in the

onset, abnormal function of the vitiated *Dosha* in the body, timing of appearance of the symptoms in a day and persistence of the symptoms for longer time. The irregularity in onset of the symptoms refers to appearance of the symptoms sometimes in the *Shiras* (head); sometimes in *Prishtha* (back) and sometimes in *Jangha* (region of hamstring muscles). The irregularity the abnormal function of vitiated *Dosha* refers to clinical presentation of cold sensation sometimes and sometimes hot sensation. The irregularity in timing of appearance of symptoms refers to the appearance of the symptoms in the morning, sometimes the appearance of the symptoms in the mid-noon, sometimes the appearance of the symptoms in the evening time or sometimes in the midnight. In other words, the clinical presentation of *Vishama Jvara* is unpredictable in the manifestation of the symptoms.

दोषोऽल्पोऽहितसंभूतो ज्वरोत्सृष्टस्य वा पुनः॥31॥

धातुमन्यतमं प्राप्य करोति विषमज्वरम्।

(सन्ततः सततोऽन्येद्युस्तृतीयकचतुर्थकौ॥)

Ma.Ni. – 2/31

A lean and emaciated person in whom the disease has recently subsided, if he indulges in improper diet and regimen, then in such individuals, the *Jvara* gets manifested, even though the *Dosha* vitiated are in milder form, get strength from *Dhatu* (*Dhatu*, *Mala* and *Kala*) and also in due course of time. Once, the *Dosha* gain strength, then only the symptoms manifest in the body. After the decrease of the strength of the *Dosha*, these *Dosha* will return to their places or become hidden in the *Dhatu*s. The dormant *Dosha* will wait for the opportunity to gain strength for getting aggravated and erupted again. Till then, *Dosha* will remain dormant without producing any symptoms deep inside the *Dhatu*s.

Samprapti of Vishama Jvara as per Sushruta:

क्षामाणां ज्वरमुक्तानां मिथ्याहारविहारिणाम्।

दोषः स्वल्पोऽपि संवृद्धो देहिनामनिलेरितः॥51॥

सततान्येद्युष्कत्र्याख्यचातुर्थान् सप्रलेपकान्।

कफस्थानविभागेन यथासंख्यं करोति हि॥52॥

अहोरात्रादहोरात्रात् स्थानात् स्थानं प्रपद्यते।

ततश्चामाशयं प्राप्य घोरं कुर्याज्ज्वरं नृणाम्॥53॥ Su.Ut. – 39/51-53

• *Sushruta* has described a different mode of *Samprapti* of *Vishama Jvara*. Due to the exposure to mild *Apathya* consumption, there will be vitiation of *Dosha*.

• The sites of *Dosha* vitiation in *Vishama Jvara* are sites of *Kapha Dosha* i.e. *Amashaya* (stomach), *Uras* (chest),

G
K

95 This may be responsible for the pursed lip breathing. This is responsible for the bypass of three major functions of nose – filtration, humidification and warming of the inspiring air. This may lead to exposure to pyrogen to the respiratory passages. In this situation of *Kamaja Jvara*, the person may have the stimulation of sympathetic nervous system which leads to the dryness of the mouth, pharynx, larynx or lower respiratory passages and alimentary canal.

96 This type of fever is manifested in an individual where in the person will have variable episodes of peripheral vasoconstriction and peripheral vasodilatation. In some other occasions, the person will have absolute normal temperature for a certain period of time. As per current understanding, the person will have intermittent type of fever. In this type of fever, the person may have some duration of symptom free period where raised body temperature comes to normal.

Kantha (throat), Shiras (head) and Sandhi (joints).

- When the vitiated Dosha remain in Amashaya, it will lead to the manifestation of Satata Jvara.
- When the vitiated Dosha remain in Uras, these will move to Amashaya in one day and thus leads to the manifestation of Anyedyushka Jvara.
- When the vitiated Dosha remain in Kantha, these will move to Uras in one day. From the Uras, these Dosha will move to Amashaya and thus leads to the manifestation of Jvara on every third day, Tritiyaka Jvara.
- When the vitiated Dosha remain in Shiras, due to further vitiation, these will move into Kantha in one day. From the Kantha, move to Uras in one day and from Uras move to Amashaya in one day. Thus, Chaturthaka Jvara will be manifested every 4th day.
- When the vitiated Dosha is observed in Sandhi, that will be responsible for the manifestation of Pralepaka Jvara.
- When the vitiated Dosha present in Amashaya, Uras, Kantha, Shiras and Sandhi simultaneously, these vitiated Dosha are responsible for the manifestation of Santata Jvara.

Lakshana of Vishama Jvara:

(स्रोतोमखिता दोषा गुरवो रसवाहिभिः।
सर्वदेहानुगाः स्तब्धा ज्वरं कुर्वन्ति सन्ततम्॥)
सप्ताहं वा दशाहं वा द्वादशाहमथापि वा।
सन्तत्या योऽविसर्गी स्यात्सन्ततः स निगद्यते॥34॥
अहोरात्रे सततको द्वौ कालाधनुवर्तते।
अन्येष्वुष्कस्त्वहोरात्रे एककालं प्रवर्तते॥35॥
तृतीयकस्तृतीयेऽह्नि, चतुर्थेऽह्नि चतुर्थकः।
केचिद्भूताभिषङ्गोत्थं ब्रुवते विषमज्वरम्॥36॥
राज्यहोः षट्सु कालेषु कीर्तितेषु यथा पुरा॥62॥
प्रसह्य विषमोऽभ्येति मानवं बहुधा ज्वरः।
स चापि विषमो देहं न कदाचिद्विमुञ्चति॥63॥
ग्लानिगौरव काशयेभ्यः स समान्न प्रमुच्यते।
वेगे तु समतिक्रान्ते गतोऽयमिति लक्ष्यते॥64॥
धात्वन्तरस्थो लीनत्वान्न सौक्ष्म्यादुपलभ्यते।
अल्पदोषेन्धनः क्षीणः क्षीणेन्धन इवानलः॥65॥

Ma.Ni. - 2/31-36

Su.Ut. - 39/62-65

- The disease is characterised by being continuous or intermittent fever with remissions for a particular period of time. It is associated with exhaustion, heaviness of the body or emaciation.

- Vishama Jvara can manifest in six different periods of the day and night. These time periods are

1. Purvahna – fore noon
2. Madhyahna – mid day
3. Aparahna – after noon
4. Pradosha – evening

5. Ardharatra - midnight

6. Pratyusha – early morning

- During these periods, the Dosha become powerful and exhibit their symptoms in various ways. As the symptoms got manifested, the virulence of the Dosha decreases. The symptoms will subside after the reduction of virulence of Dosha and become dormant in the body and stay for the longer time and will not be eliminated from the body. This results in the presenting complaints of exhaustion, heaviness of the body and emaciation during the episode of the illness. The disease Jvara remains latent in the body of the patient deep within the Dhātu just like minute form of spark of fire with little fuel.

Types of Vishama Jvara: 5 types of Vishama Jvara.

1. Santata Jvara
2. Satata Jvara
3. Anyedyushka Jvara
4. Tritiyaka Jvara
5. Chaturthaka Jvara

Santata Jvara:

स्रोतोभिर्विसृता दोषा गुरवो रसवाहिभिः॥53॥
सर्वदेहानुगाः स्तब्धा ज्वरं कुर्वन्ति सन्ततम्।
दशाहं द्वादशाहं वा सप्ताहं वा सुदुःसहः॥54॥
स शीघ्रं शीघ्रकारित्वात् प्रशमं याति हन्ति वा।
कालदृष्यप्रकृतिभिर्दोषस्तुल्यो हि सन्ततम्॥55॥
निष्प्रत्यनीकः कुरुते तस्माज्जेयः सुदुःसहः।
यथा धातूस्तथा मूत्रं पुरीषं चानिलादयः॥56॥
युगपच्छानुपद्यन्ते नियमात् सन्तते ज्वरे।
स शुच्या वाऽप्यशुद्ध्या वा रसादीनामशेषतः॥57॥
सप्ताहादिषु कालेषु प्रशमं याति हन्ति वा।
यदा तु नातिशुध्यन्ति न वा शुध्यन्ति सर्वशः॥58॥
द्वादशैते समुद्दिष्टाः सन्ततस्याश्रयास्तदा।
विसर्गे द्वादशे कृत्वा दिवसेऽव्यक्तलक्षणम्॥59॥
दुर्लभोपशमः कालं दीर्घमप्यनुवर्तते।
इति बुवा ज्वरं वैद्य उपक्रमेत्तु सन्ततम्॥60॥

Ca.Ci. - 3/53-60

- The Doshas get vitiated which are Guru (heavy to digest) will spread all over the body through Rasavaha Srotas. Due to its heaviness, it may get entangled anywhere in the Srotas. This stagnation produces Santata Jvara. The nature of Jvara is continuous in form which stays for the longer period.
- The nature of fever is Shighrakariva (acute in form) and gets cured or dies in 7 or 10 or 12 days of onset.
- Santata Jvara is Nishpratyanika in nature. The term Pratyatika refers to antagonistic and Nishpratyanika refers not antagonistic. That means the Kala, Prakriti, Dushya and vitiated Dosha work coherently in the manifestation of Santata Jvara. For example, the

person of *Kapha Prakriti* developing the symptoms in morning time involving *Rasa Dhatu* indicates more severity of the *Santata Jvara*.

- The durations of *Vata* dominant *Santata Jvara* is 7 days, *Pitta* dominant *Santata Jvara* is 10 days and *Kapha* dominant *Santata Jvara* is 12 days. Even after 12 days of appearance of *Jvara*, symptoms may persist even after the subsidence of vitiated *Dosha*. the refore the physician should plan to initiate the treatment otherwise it may turn difficult to cure.
- Various modalities of treatment are explained, but *Apatarpana* should be the first line of treatment.

Satata Jvara:

रक्तधात्वाधयः प्रायो दोषः सततकं ज्वरम्॥61॥

सप्रत्यनीकः कुरुते कालवृद्धिक्षयात्मकम्।

अहोरात्रे सततको द्वौ कालावनुवर्तते॥62॥

Ca.Ci. - 3/61-62

- *Satata Jvara* is manifested if the vitiated *Dosha* gets lodged in *Rakta Dhatu*. This usually counteracts with *Kala* with respect to increase and decrease in intensity.
- Therefore, depending upon the *Kala*, there will be aggravation and remission from *Jvara*.
- *Satata Jvara* rises twice in a day.

Anyedyushka Jvara:

कालप्रकृतिदृष्ट्याणां प्रायैवान्यतमाद्बलम्।

अन्येद्युष्कं ज्वरं दोषो रुवा मेदोवहाः सिराः॥63॥

सप्रत्यनीको जनयत्येककालमर्हति।

सोऽन्येद्युः पिशिताश्रितः॥

Ca.Ci. - 3/63-64

Ma.Ni. - 2/32

- Depending upon time, individual constitution and status of *Dushya*, there will be aggravation of symptoms.
- Vitiated *Dosha* will cause the affliction of *Medovaha Srotas*. As per *Vagbhata*, there will be the vitiation of *Mamsa Dhatu*.
- The fever manifests in once in a day.

Tritiyaka Jvara:

दोषोऽस्थिमज्जगः कुर्यात्तृतीयकचतुर्थकौ॥64॥

मेदोगतस्तृतीयेऽह्नि,

तृतीयकस्तृतीयेऽह्नि,

Ca.Ci. - 3/64

Ma.Ni. - 2/33

Ma.Ni. - 2/36

- *Tritiyaka Jvara* is manifested due to the involvement of *Asthi* and *Majja Dhatu* as per *Charaka*. As per *Madhavakara*, *Medas* is involved in *Tritiyaka Jvara*.
- In *Tritiyaka Jvara*, *Jvara* is manifested in every 3rd day.

Types of Tritiyaka Jvara

कफपित्तात्रिकग्राही पृष्ठाद्वातकफात्मकः।

वातपित्ताच्छिरोग्राही त्रिविधः स्यात्तृतीयकः॥37॥

Ma.Ni. 2/37

3 types of *Tritiyaka Jvara* are

1. *Kaphapittaja* – *Trika* or sacral region is afflicted.
2. *Vatakapaja* – *Prishtha* or back is afflicted.
3. *Vatapittaja* – *Shiras* or head is afflicted.

Chaturthaka Jvara:

दोषोऽस्थिमज्जगः कुर्यात्तृतीयकचतुर्थकौ॥64॥

Ca.Ci. - 3/64

त्वस्थिमज्जगतः पुनः।

कुर्याच्चतुर्थकं घोरमन्तकं

Ma.Ni. - 2/32

चतुर्थेऽह्नि चतुर्थकः।

Ma.Ni. - 2/36

- As per *Charaka* and *Madhavakara*, *Chaturthaka Jvara* is manifested due to the involvement of *Asthi* and *Majja Dhatu*.
- In *Chaturthaka Jvara*, *Jvara* is manifested in every 4th day.

Types of Chaturthaka Jvara:

चतुर्थको दर्शयति प्रभावं द्विविधं ज्वरः।

जङ्घाभ्यां श्लैष्मिकः पूर्व शिरस्तोऽनिलसंभवः॥38॥

Ma.Ni. - 2/38

2 types of *Chaturthaka Jvara* are

1. *Kapha dominant Chaturthaka Jvara* – where *Jangha* or hamstrings are involved.
2. *Vata dominant Chaturthaka Jvara* – where *Shiras* or head is involved.

Chaturthaka Viparyaya Jvara:

विषमज्वर एवान्यश्चतुर्थकविपर्ययः।

स मध्ये ज्वरयत्यही आदावन्ते च मुञ्चति॥39॥

Ma.Ni. - 2/39

This is a variant of *Vishama Jvara*. In this type of *Jvara*, *Jvara* will be manifested in 2nd and 3rd day. During 1st and 4th days, the person will be free from the symptoms of *Jvara*.

Punaravartaka Jvara: *Punaravartaka Jvara* is a type of *Jvara* which recurs in the individual after exposure to certain etiological factors.

Nidana of Punaravartaka⁹⁷ Jvara:

सज्वरो ज्वरमुक्तश्च विदाहीनि गुरुणि च।

असात्म्यान्त्यन्नपानानि विरुद्धानि च वर्जयेत्॥330॥

व्यवायमतिचेष्टाश्च स्नानमत्यशनानि च।

तथा ज्वरः शमं याति प्रशान्तो जायते न च॥331॥

व्यायामं च व्यवायं च स्नानं चक्रमणानि च।

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- 97 This can be attributable to intermittent type of fever or recurrent fever or irregular fever. Let us take one example of *Vyayama* (exercise). In this situation, the person may have increased blood supply to the area where more muscular activity is seen. In that condition, pyrogen which may be restricted to one place may move to the part of the body which receives more amount of blood supply. Thus, there may be a possibility of antigen antibody reaction which may be the reason for raised body temperature. Likewise, you can give explanations to other causes of *Punaravartaka Jvara*.

ज्वरमुक्तो न सेवेत यावन्न बलवान् भवेत्॥३३२॥ Ca.Ci. - 3/330-332
When the person is getting relief from *Jvara* for a short period of time, if he indulges in.

- **Vidahi Ahara** – consumption of food substances responsible for the burning sensation or thirst etc.
- **Guru Ahara** – consumption of food substances which are heavy to digest.
- **Asatmya Annapana** – consumption of unwholesome food substances.
- **Viruddha Ahara** – consumption of incompatible food substances.
- **Ativyavaya** – excess indulgence in sexual activities.
- **Ati cheshta** – excess activities.
- **Snana** – bathing.
- **Atyashana** – consumption of large amount of food substances at a time.
- **Vyayama** – performing excess exercises.
- **Chankramana** – excess walking.

Lakshana of Punaravartaka Jvara:

असंजातबलो यस्तु ज्वरमुक्तो निषेवते।
वर्ज्यमेतन्नरस्तस्य पुनरावर्तते ज्वरः॥३३३॥
दुर्हतेषु च दोषेषु यस्य वा विनिवर्तते।
स्वल्पेनाप्यपचारेण तस्य व्यावर्तते पुनः॥३३४॥
चिरकालपरिविलष्टं दुर्बलं हीनतेजसम्।
अचिरेणैव कालेन स हन्ति पुनरागतः॥३३५॥
अथवाऽपि परीपाकं धातुष्वेव क्रामन्मलाः।
यान्ति ज्वरमकुर्वन्तस्ते तथाऽप्यपकुर्वन्ते॥३३६॥
दीनतां श्वयथुं ग्लानिं पाण्डुतां नात्रकामताम्।
कण्डूत्कोठपिडकाः कुर्वन्त्यग्निं च ते मृदुम्॥३३७॥
एवमन्येऽपि च गदा व्यावर्तन्ते पुनर्गताः।
अनिर्घातेन दोषाणामल्पैरप्यहितैर्नृणाम्॥३३८॥
निर्वृत्तेऽपि ज्वरे तस्माद्यथावस्थं यथाबलम्।
यथाप्राणं हरेद्दोषं प्रयोगैर्वा शमं नयेत्॥३३९॥ Ca.Ci. - 3/333-339

- Due to subsidence of fever and the incomplete recovery from the illness and the person is indulging in above said etiological factors even for a shorter duration, there will be recurrence in the manifestation of *Jvara*.
- The person will be presented with clinical features such as *Dinata* (illlook), *Shvayathu* (generalised or localised swelling), *Glani* (exhaustion or fatigue), *Panduta* (anemia), *Anannakamata* (lack of desire to consume food), *Kandu* (itching), *Utkotha* (wheals), *Pidaka* (boils) and *Agnimandya* (loss of appetite) and other diseases.

Dhatugata Jvara

The term *Dhatugata Jvara* refers to the presence of vitiated *Dosha* in the *Dhatu* in the body. The severity of the

Lakshana is becoming more and more severe depending upon the involvement of *Dhatu*. The following are the *Lakshana* observed in the different *Dhatugata Jvara*.

गुरुत्वं दैन्यमुद्वेगः सदनं छद्यरोचकौ।
रसस्थिते बहिस्तापः साङ्गमर्दौ विजृम्भणम्॥७६॥
रक्तोष्णाः पिडकास्तृष्णा सरक्तं धीवनं मुहुः।
दाहारागभ्रममदप्रलापा रक्तसंस्थिते॥७७॥
अन्तर्दाहः सतृष्णमोहः सग्लानिः सृष्टविकता।
दौर्गन्ध्यं गात्रविक्षेपो ज्वरे मांसस्थिते भवेत्॥७८॥
स्वेदस्तीवा पिपासा च प्रलापो वम्यभीक्षणशः।
स्वगन्धस्यासहत्वं च मेदःस्थे ग्लान्यरोचकौ॥७९॥
विरेकघमने चोभे सास्थिभेदं प्रकूजनम्।
विक्षेपणं च गात्राणां श्वासश्चास्थिगते ज्वरे॥८०॥
हिक्का श्वासस्तथा कासस्तमसश्चातिदर्शनम्।
मर्मच्छेदो वहिः शैत्यं दाहोऽन्तश्चैव मज्जगे॥८१॥
शुक्रस्थानगतः शुक्रमोक्षं कृत्वा विनाश्य च।
प्राणं वाय्वग्निसोमैश्च सार्धं गच्छत्यसौ विभुः॥८२॥
रसरक्ताश्रितः साध्यो मेदोमांसगतश्च यः।
अस्थिमज्जगतः कृच्छ्रः शुक्रस्थो नैव सिद्ध्यति॥८३॥

Ca.Ci. - 2/76-83

Lakshana of Rasadhatugata Jvara: *Gurutva* (heaviness of the body), *Dainya* (ill look), *Udvega* (anxiety), *Sadana* (sluggish movement of the body parts), *Chardi* (vomiting), *Arochaka* (tastelessness), *Bahistapa* (raised temperature in the exterior or skin), *Angamarda* (bodyache), *Vijrimbhana* (excessive yawning).

Lakshana of Raktadhatugata Jvara: *Rakta* and *Ushna* *Pidaka* (reddish and warm rashes), *Trishna* (thirst), *Muhuh Sarakta* *Shtivana* (frequent expectoration of sputum mixed with blood), *Daha* (burning sensation), *Raga* (redness), *Bhrama* (giddiness), *Mada* (intoxication or altered sensation), *Pralapa* (irrelevant or incomprehensible speech).

Lakshana of Mamsadhatugata Jvara: *Antardaha* (burning sensation felt inside the body), *Trishna* (thirst), *Moha* (confusion), *Glani* (exhaustion), *Shrishta Vit* (unformed stools), *Daurgandhya* (foul smell from the body), *Gatravikshepa* (involuntary movements of the body parts).

Lakshana of Medodhatugata Jvara: *Sveda* (sweating), *Tivra Pipasa* (severe thirst), *Pralapa* (irrelevant speech), *Abhikshna* *Vamana* (projectile or effortless vomiting), *Svagandhasya* *Asahatva* (aversion towards smell of one's own body), *Glani* (exhaustion), *Arochaka* (tastelessness).

Lakshana of Asthidhatugata Jvara: *Virechana* and *Vamana* at a time (vomiting and diarrhoea at a time), *Asthibheda* (tearing type of pain in the bones), *Prakujana* (specific type of sound made by the patient), *Gatranam Vikshepa* (involuntary movements of the body parts), *Shvasa* (breathlessness).

Lakshana of Majjadhatugata Jvara: *Hikka* (hiccup), *Shvasa* (breathlessness), *Kasa* (cough), *Tamasah Darshana* (darkness in front of the eyes), *Marmaccheda* (tearing type of pain in vital points), *Bahih Shaitya* (feeling of coldness).

in the skin or exterior), *Antardaha* (burning sensation inside the body).

Lakshana of Shukradhatugata Jvara: *Vinasha* of the individual after *Shukra Moksha* (destruction of the body after ejaculation of semen), *Marana* (death).

Sadhysadhyata of Dhatugata Jvara:

- ⊗ *Rasadhatugata*, *Raktadhatugata*, *Mamsadhatugata* and *Medodhatugata Jvara* are considered as *Sadhya* (easily curable) in nature.
- ⊗ *Asthidhatugata* and *Majjadhatugata Jvara* are considered as *Kricchrasadhyata* (difficult to cure) in nature.
- ⊗ *Shukradhatugata Jvara* is considered as *Asadhyata* (incurable) in nature.

Sannipataja Jvara:

सन्निपातज्वरस्योर्ध्वं त्रयोदशविधस्य हि।
 प्राक्सूत्रितस्य वक्ष्यामि लक्षणं वै पृथक् पृथक्॥१०॥
 भ्रमः पिपासा दाहश्च गौरवं शिरसोऽतिरूक्।
 वातपित्तलब्धे विद्यालिङ्गं मन्दकफे ज्वरे॥११॥
 शैत्यं कासोऽरुचिस्तन्द्रापिपासादाहरुच्यथाः।
 वातश्लेष्मोलब्धे व्याधौ लिङ्गं पित्तावरे विदुः॥१२॥
 छर्दिः शैत्यं मुहुर्दाहस्तृष्णा मोहोऽस्थिवेदना।
 मन्दवाते व्यवस्यन्ति लिङ्गं पित्तकफोलब्धे॥१३॥
 सन्ध्यस्थिशिरसः शूलं प्रलापो गौरवं भ्रमः।
 वातोलब्धे स्याद् द्युनो तृष्णा कण्ठास्यशुष्कता १४
 रक्तविण्मूत्रता दाहः खेदस्तुद् बलसंक्षयः।
 मूर्च्छा चेति त्रिदोषे स्याल्लिङ्गं पित्ते गरयसि॥१५॥
 आलस्यारुचिहृल्लासदाहवम्यरतिभ्रमैः।
 कफोलब्धं सन्निपातं तन्द्राकासेन चादिशेत्॥१६॥
 प्रतिश्या छर्दिरालस्यं तन्द्राऽरुच्यग्निमार्दवम्।
 हीनवाते पित्तमध्ये लिङ्गं श्लेष्माधिके मतम्॥१७॥
 हरिद्रमूत्रनेत्रत्वं दाहस्तृष्णा भ्रमोऽरुचिः।
 हीनवाते मध्यकफे लिङ्गं पित्ताधिके मतम्॥१८॥
 शिरोरुग्नेपथुः श्वासः प्रलापश्छर्दरोचकौ।
 हीनपित्ते मध्यकफे लिङ्गं स्यान्मरुताधिके॥१९॥
 शीतको गौरवं तन्द्रा प्रलापोऽस्थिशिरसतिरूक्।
 हीनपित्ते वातमध्ये लिङ्गं श्लेष्माधिके विदुः॥१००॥
 श्वासः कासः प्रतिश्यायो मुखशोषोऽतिपाश्चर्यकृ।
 कफहीने पित्तमध्ये लिङ्गं वाताधिके मतम्॥१०१॥
 बचभेदोऽग्निदौर्बल्यं तृष्णा दाहोऽचिर्भ्रमः।
 कफहीने वातमध्ये लिङ्गं पित्ताधिके विदुः॥१०२॥

Vata Pittolbana Manda Kaphaja Jvara: This is associated with giddiness, thirst, burning sensation, heaviness and excessive headache.

Vata Kapholbana Manda Pittaja Jvara: This is associated with coldness, cough, anorexia, drowsiness, thirst, burning sensation, pain and distress.

Pitta Kapholbana Manda Vataja Jvara: This is associated

with vomiting, burning sensation and coldish sensation alternatively, thirst, unconsciousness and pain in bones.

Vatolbana Manda Pitta Kaphaja Jvara: This is associated with pain in joints, bones and head; delirium, heaviness, giddiness, thirst, dryness of throat and mouth.

Pittolbana Manda Vata Kaphaja Jvara: This is associated with reddish discolouration of stool and urine; burning sensation, sweating, thirst, loss of strength and fainting.

Kapholbana Manda Vata Pittaja Jvara: This is associated with laziness, anorexia, nausea, burning sensation, vomiting, restlessness, giddiness, drowsiness and cough.

Kapholbana Madhya Pitta and Manda Vataja Jvara: This is associated with coryza, vomiting, laziness, drowsiness, anorexia, loss of function of *Agni*.

Pittolbana, Madhya Kapha and Manda Vataja Jvara: This is associated with predominance of *Pitta* followed by moderate *Kapha* increase and mild vitiation of *Vata*, yellowish discolouration of urine and eyes; burning sensation, thirst, giddiness and anorexia.

Vatolbana, Madhya Kapha and Manda Pittaja Jvara: This is associated with predominance of *Vata* followed by moderate increase of *Kapha* and mild vitiation of *Pitta*; headache, tremors, breathlessness, delirium, vomiting and anorexia.

Kapholbana, Madhya Vata and Manda Pittaja Jvara: This is associated with predominance of *Kapha*, followed by moderate increase of *Vata* and mild vitiation of *Pitta*, cold sensation of body parts, heaviness, drowsiness, delirium and excessive pain in bone and head.

Vatolbana Madhya Pitta and Manda Kaphaja Jvara: This is associated with predominance of *Vata*, followed by moderate increase of *Pitta* and mild vitiation of *Kapha*; breathlessness, cough, coryza, dryness of mouth and excess chest discomfort.

Pittolbana, Madhya Vata and Manda Kaphaja Jvara: This is associated with predominance of *Pitta*, followed by moderate increase of *Vata* and mild vitiation of *Kapha*, *diarrhoea*, sluggish digestion, thirst, burning sensation, anorexia and giddiness.

Sama Sannipataja Jvara:

क्षणे दाहः क्षणे शीतमस्थिसन्ध्यशिरोरुजा॥१०३॥
 सास्त्रावे कलुषे रक्ते निर्भुग्ने चापि दर्शने।
 सस्वनी सरुजी कणौ कण्ठः शुक्लैरिववृतः॥१०४॥
 तन्द्रा मोहः प्रलापश्च कासः श्वासोऽरुचिर्भ्रमः।
 परिदग्धा खरस्पर्श जिह्वा खस्ताङ्गता परम्॥१०५॥
 शीवनं रक्तपित्तस्य कफेनोन्मिथितस्य च।
 शिरसो लोठनं तृष्णा निद्रानाशो हृदि व्यथा॥१०६॥
 स्वेदमूत्रपुरीषाणां चिराद्दर्शनमल्पशः।
 शत्वं नातिगात्राणां प्रततं कण्ठकूजनम्॥१०७॥
 कोठानां श्यावरतानां मण्डलानां च दर्शनम्।
 मूकत्वं स्रोतसां पाको गुरुत्वमुदरस्य च॥१०८॥

त्रिरात् पाकश्च दोषाणां सन्निपातज्वराकृतिः।

Ca.Ci. – 3/103-108

- Frequent **appearance**⁹⁸ of burning sensation and cold sensation in the body.
- Discomfort in bones, joints and head.
- Eyes are filled with tears, muddy discolouration, reddish discolouration, sunken eyes.
- Ringing sensation or tinnitus and pain in the **ears**⁹⁹.
- Sensation of filling of throat with **Shuka**¹⁰⁰ (paddy like substance having pointed ends on either sides).
- Drowsiness, altered state of consciousness, irrelevant and incomprehensible **speech**¹⁰¹.
- Cough, **breathlessness**¹⁰².
- **Tastelessness**¹⁰³.
- **Giddiness**¹⁰⁴.
- **Rough tongue**¹⁰⁵ and sensation of roughness of the tongue.
- Dislocation of the body parts or joints.
- Presence of blood in the sputum or presence of mucus along with blood in the **sputum**¹⁰⁶.

- Involuntary or uncontrolled movement of head.
- **Thirst**¹⁰⁷.
- **Loss of sleep**¹⁰⁸.
- Discomfort in the **chest region**¹⁰⁹.
- Very little or delayed appearance of sweat, urine and **faecal matter**¹¹⁰.
- **Emaciation of the body**¹¹¹.
- Continuous production of sound from throat.
- Appearance of circular blackish or reddish rashes.
- Dumbness.
- **Suppuration**¹¹² in various channels of the body.
- Heaviness of the abdominal region.
- Delayed process of *Doshapaka* in the body.

8 types of Jvara

Vataja Jvara:

वेपथुर्विषमो वेगः कण्ठौष्ठपरिशोषणम्।

- 98 This is attributable to the frequent peripheral vaso dilatation and vaso constriction. When the person is having peripheral vaso dilatation, there may be a possibility of burning sensation and when the person is having peripheral vaso constriction, the person may present with cold sensation. This presenting symptom indicates unstable pathophysiology in terms of peripheral circulatory abnormalities.
- 99 This is attributable to poor peripheral circulation or to nasal blockage which obstruct the eustachian tube. When there is poor peripheral circulation, there will be less formation of perilymph and endolymph and thus the person may be presented with hearing impairments. When the person is having nasal blockage, this will block the eustachian tube and this is responsible for the pain in the ears.
- 100 This is the Purvarupa (premonitory symptom) observed in the patient of Kasa. This is indicative of respiratory dysfunction.
- 101 This is attributable to poor circulation over brain or nervous system. This symptom can be seen in case of syncopal attack which is a sequel of hypovolemia.
- 102 This symptom is attributable to abnormal respiratory function.
- 103 This is attributable to the poor alimentary canal function. This is seen in sympathetic nervous system activation.
- 104 This is attributable to the poor circulation to the head and neck. This is occurring due to the syncopal attack.
- 105 This is attributable to poor alimentary canal function. This occurs as a result of activation of sympathetic nervous system which in turn results in poor secretion in alimentary canal including salivary glands. That is the reason, the person may have roughness of the tongue as a sequel of poor metabolism in the body.
- 106 The presence of blood in the sputum may indicate serious respiratory infection such as tuberculosis or rupture of the capillaries in the respiratory system.
- 107 The presence of thirst implies that decrease of the fluid in the intravascular, interstitial and intracellular compartments. But, when an individual consumes considerable amount of water, only after 30 to 60 minutes, he may be in a position to have relief from the thirst. This may indicate loss of the fluid most in the form of sweat or through breath. The person may also have the habit of breathing through open mouth i.e. panting process. This is a quick mechanism where thirst may appear in very short time due to mouth breathing.
- 108 There are two types of sleep, they are REM sleep (25% of the total duration of sleep) and NREM sleep (75% of the total duration of sleep). REM sleep recurs every 90 minutes where REM may have the time duration of 5 to 30 minutes.
 - In this condition, loss of sleep can be attributable to increased time duration of REM sleep and decreased time duration of NREM sleep.
- 109 As per current understanding, the presence of fever considered as constitutional symptom. The association of the symptom determines the systemic dysfunction. When an individual presented with chest discomfort, it implies the affliction of either respiratory system or cardiovascular system or alimentary canal or Musculo skeletal system or neurological system. The location of the symptom, radiation of the symptom, presence of associated symptoms. For example, if the person is presented with severe degree of cough with sputum, may have chest discomfort on either sides of the chest.
- 110 Very little formation of sweat and urine implies decreased hydration of the body secondary to the fluid loss naturally occurs as a result of increased heat in the body. In this, there will be shrinkage of the intravascular compartment. In case of little faecal matter formation, there may be activation of sympathetic nervous system which in turn leads to contraction of anal sphincter which leads to absorption of 75% of the water portion from the faecal matter. Because of this, the person may have the complaint of constipation.
- 111 When an individual is presented with severe form of fever, that is caused by two pathological mechanisms viz. increased heat production or decreased heat loss from the body. In case of increased heat production, there will be enhanced catabolism in the body, that is more utilisation of the nutrients either circulating in the blood or body fluid or deposited in the body tissues. This will lead to the emaciation of the body.
- 112 When there is suppuration happening in the body, there will be interaction between the white blood cells with pathogens in the body. This mechanism is responsible for the manifestation of fever.

in the skin or exterior), *Antardaha* (burning sensation inside the body).

Lakshana of Shukradhatugata Jvara: *Vinasha* of the individual after *Shukra Moksha* (destruction of the body after ejaculation of semen), *Marana* (death).

Sadhysadhyata of Dhatugata Jvara:

- *Rasadhatugata*, *Raktadhatugata*, *Mamsadhatugata* and *Medodhatugata Jvara* are considered as *Sadhya* (easily curable) in nature.
- *Asthidhatugata* and *Majjadhatugata Jvara* are considered as *Kricchrasadhya* (difficult to cure) in nature.
- *Shukradhatugata Jvara* is considered as *Asadhya* (incurable) in nature.

Sannipataja Jvara:

सन्निपातज्वरस्योर्ध्वं त्रयोदशविधस्य हि।
 प्राक्सूत्रितस्य वक्ष्यामि लक्षणं वै पृथक् पृथक्॥१०॥
 भ्रमः पिपासा दाहश्च गौरवं शिरसोऽतिरुक्।
 वातपित्तल्बणे विद्यालिङ्गं मन्दकफे ज्वरे॥११॥
 शैत्यं कासोऽरुचिस्तन्द्रापिपासादाहरुयथाः।
 वातश्लेष्मोल्बणे व्याधौ लिङ्गं पित्तावरे विदुः॥१२॥
 छर्दिः शैत्यं मुहुर्दाहस्तृष्णा मोहोऽस्थिवेदना।
 मन्दवाते व्यवस्यन्ति लिङ्गं पित्तकफोल्बणे॥१३॥
 सन्ध्यस्थिशिरसः शूलं प्रलापो गौरवं भ्रमः।
 वातोल्बणे स्याद् द्युनो तृष्णा कण्ठास्यशुष्कता १४
 रक्तविण्मूत्रता दाहः खेदस्तुब्ध बलसंक्षयः।
 मूर्च्छा चेति त्रिदोषे स्याल्लिङ्गं पित्ते गरयसि॥१५॥
 आलस्यारुचिहृल्लासदाहवम्यरतिभ्रमैः।
 कफोल्बणं सन्निपातं तन्द्राकासेन चादिशेत्॥१६॥
 प्रतिश्या छर्दिरालस्यं तन्द्राऽरुच्यग्निमार्दवम्।
 हीनवाते पित्तमध्ये लिङ्गं श्लेष्माधिके मतम्॥१७॥
 हारिद्रमूत्रनेत्रत्वं दाहस्तृष्णा भ्रमोऽरुचिः।
 हीनयाते मध्यकफे लिङ्गं पित्ताधिके मतम्॥१८॥
 शिरोरुवेपथुः श्वासः प्रलापश्छर्दरोचकौ।
 हीनपित्ते मध्यकफे लिङ्गं स्यान्मारुताधिके॥१९॥
 शीतको गौरवं तन्द्रा प्रलापोऽस्थिशिरसतिरुक्।
 हीनपित्ते वातमध्ये लिङ्गं श्लेष्माधिके विदुः॥२०॥
 श्वासः कासः प्रतिश्यायो मुखशोषोऽतिपाश्वरुक्।
 कफहीने पित्तमध्ये लिङ्गं वाताधिके मतम्॥२१॥
 बचभेदोऽग्निदौर्बल्यं तृष्णा दाहोऽचिर्भ्रमः।
 कफहीने वातमध्ये लिङ्गं पित्ताधिके विदुः॥२२॥

Vata Pittolbana Manda Kaphaja Jvara: This is associated with giddiness, thirst, burning sensation, heaviness and excessive headache.

Vata Kapholbana Manda Pittaja Jvara: This is associated with coldness, cough, anorexia, drowsiness, thirst, burning sensation, pain and distress.

Pitta Kapholbana Manda Vataja Jvara: This is associated

with vomiting, burning sensation and coldish sensation alternatively, thirst, unconsciousness and pain in bones.

Vatolbana Manda Pitta Kaphaja Jvara: This is associated with pain in joints, bones and head; delirium, heaviness, giddiness, thirst, dryness of throat and mouth.

Pittolbana Manda Vata Kaphaja Jvara: This is associated with reddish discolouration of stool and urine; burning sensation, sweating, thirst, loss of strength and fainting.

Kapholbana Manda Vata Pittaja Jvara: This is associated with laziness, anorexia, nausea, burning sensation, vomiting, restlessness, giddiness, drowsiness and cough.

Kapholbana Madhya Pitta and Manda Vataja Jvara: This is associated with coryza, vomiting, laziness, drowsiness, anorexia, loss of function of *Agni*.

Pittolbana, Madhya Kapha and Manda Vataja Jvara: This is associated with predominance of *Pitta* followed by moderate *Kapha* increase and mild vitiation of *Vata*, yellowish discolouration of urine and eyes; burning sensation, thirst, giddiness and anorexia.

Vatolbana, Madhya Kapha and Manda Pittaja Jvara: This is associated with predominance of *Vata* followed by moderate increase of *Kapha* and mild vitiation of *Pitta*; headache, tremors, breathlessness, delirium, vomiting and anorexia.

Kapholbana, Madhya Vata and Manda Pittaja Jvara: This is associated with predominance of *Kapha*, followed by moderate increase of *Vata* and mild vitiation of *Pitta*, cold sensation of body parts, heaviness, drowsiness, delirium and excessive pain in bone and head.

Vatolbana Madhya Pitta and Manda Kaphaja Jvara: This is associated with predominance of *Vata*, followed by moderate increase of *Pitta* and mild vitiation of *Kapha*; breathlessness, cough, coryza, dryness of mouth and excess chest discomfort.

Pittolbana, Madhya Vata and Manda Kaphaja Jvara: This is associated with predominance of *Pitta*, followed by moderate increase of *Vata* and mild vitiation of *Kapha*, *diarrhoea*, sluggish digestion, thirst, burning sensation, anorexia and giddiness.

Sama Sannipataja Jvara:

क्षणे दाहः क्षणे शीतमस्थिसन्धिशिरोरुजा॥१०३॥
 सास्त्रावे कलुषे रक्ते निर्भुग्ने चापि दर्शने।
 सस्वनी सरुजी कर्णौ कण्ठः शुक्लैरिवावृतः॥१०४॥
 तन्द्रा मोहः प्रलापश्च कासः श्वासोऽरुचिर्भ्रमः।
 परिदग्धा खरस्पर्श जिहा स्त्रस्ताङ्गता परम्॥१०५॥
 धीवनं रक्तपित्तस्य कफेनोन्मिथितस्य च।
 शिरसो लोठनं तृष्णा निद्रानाशो हृदि व्यथा॥१०६॥
 स्वेदमूत्रपुरीषाणां चिराद्दर्शनमल्पशः।
 शत्वं नातिगात्राणां प्रततं कण्ठकूजनम्॥१०७॥
 कोठानां श्यावरतानां मण्डलानां च दर्शनम्।
 मूकत्वं स्रोतसां पाको गुरुत्वमुदरस्य च॥१०८॥

विरात् पाकश्च दोषाणां सन्निपातज्वराकृतिः।

Ca.Ci. - 3/103-108

- Frequent **appearance⁹⁸** of burning sensation and cold sensation in the body.
- Discomfort in bones, joints and head.
- Eyes are filled with tears, muddy discolouration, reddish discolouration, sunken eyes.
- Ringing sensation or tinnitus and pain in the **ears⁹⁹**.
- Sensation of filling of throat with **Shuka¹⁰⁰** (paddy like substance having pointed ends on either sides).
- Drowsiness, altered state of consciousness, irrelevant and incomprehensible **speech¹⁰¹**.
- Cough, **breathlessness¹⁰²**.
- **Tastelessness¹⁰³**.
- **Giddiness¹⁰⁴**.
- **Rough tongue¹⁰⁵** and sensation of roughness of the tongue.
- Dislocation of the body parts or joints.
- Presence of blood in the sputum or presence of mucus along with blood in the **sputum¹⁰⁶**.

- Involuntary or uncontrolled movement of head.
- **Thirst¹⁰⁷**.
- **Loss of sleep¹⁰⁸**.
- Discomfort in the **chest region¹⁰⁹**.
- Very little or delayed appearance of sweat, urine and **faecal matter¹¹⁰**.
- **Emaciation of the body¹¹¹**.
- Continuous production of sound from throat.
- Appearance of circular blackish or reddish rashes.
- Dumbness.
- **Suppuration¹¹²** in various channels of the body.
- Heaviness of the abdominal region.
- Delayed process of *Doshapaka* in the body.

8 types of Jvara

Vataja Jvara:

वेपथुर्विषमो वेगः कण्ठौष्ठपरिशोषणम्।

- 98 This is attributable to the frequent peripheral vaso dilatation and vaso constriction. When the person is having peripheral vaso dilatation, there may be a possibility of burning sensation and when the person is having peripheral vaso constriction, the person may present with cold sensation. This presenting symptom indicates unstable pathophysiology in terms of peripheral circulatory abnormalities.
- 99 This is attributable to poor peripheral circulation or to nasal blockage which obstruct the eustachian tube. When there is poor peripheral circulation, there will be less formation of perilymph and endolymph and thus the person may be presented with hearing impairments. When the person is having nasal blockage, this will block the eustachian tube and this is responsible for the pain in the ears.
- 100 This is the Purvarupa (premonitory symptom) observed in the patient of Kasa. This is indicative of respiratory dysfunction.
- 101 This is attributable to poor circulation over brain or nervous system. This symptom can be seen in case of syncopal attack which is a sequel of hypovolemia.
- 102 This symptom is attributable to abnormal respiratory function.
- 103 This is attributable to the poor alimentary canal function. This is seen in sympathetic nervous system activation.
- 104 This is attributable to the poor circulation to the head and neck. This is occurring due to the syncopal attack.
- 105 This is attributable to poor alimentary canal function. This occurs as a result of activation of sympathetic nervous system which in turn results in poor secretion in alimentary canal including salivary glands. That is the reason, the person may have roughness of the tongue as a sequel of poor metabolism in the body.
- 106 The presence of blood in the sputum may indicate serious respiratory infection such as tuberculosis or rupture of the capillaries in the respiratory system.
- 107 The presence of thirst implies that decrease of the fluid in the intravascular, interstitial and intracellular compartments. But, when an individual consumes considerable amount of water, only after 30 to 60 minutes, he may be in a position to have relief from the thirst. This may indicate loss of the fluid most in the form of sweat or through breath. The person may also have the habit of breathing through open mouth i.e. panting process. This is a quick mechanism where thirst may appear in very short time due to mouth breathing.
- 108 There are two types of sleep, they are REM sleep (25% of the total duration of sleep) and NREM sleep (75% of the total duration of sleep). REM sleep recurs every 90 minutes where REM may have the time duration of 5 to 30 minutes.
 - In this condition, loss of sleep can be attributable to increased time duration of REM sleep and decreased time duration of NREM sleep.
- 109 As per current understanding, the presence of fever considered as constitutional symptom. The association of the symptom determines the systemic dysfunction. When an individual presented with chest discomfort, it implies the affliction of either respiratory system or cardiovascular system or alimentary canal or Musculo skeletal system or neurological system. The location of the symptom, radiation of the symptom, presence of associated symptoms. For example, if the person is presented with severe degree of cough with sputum, may have chest discomfort on either sides of the chest.
- 110 Very little formation of sweat and urine implies decreased hydration of the body secondary to the fluid loss naturally occurs as a result of increased heat in the body. In this, there will be shrinkage of the intravascular compartment. In case of little faecal matter formation, there may be activation of sympathetic nervous system which in turn leads to contraction of anal sphincter which leads to absorption of 75% of the water portion from the faecal matter. Because of this, the person may have the complaint of constipation.
- 111 When an individual is presented with severe form of fever, that is caused by two pathological mechanisms viz. increased heat production or decreased heat loss from the body. In case of increased heat production, there will be enhanced catabolism in the body, that is more utilisation of the nutrients either circulating in the blood or body fluid or deposited in the body tissues. This will lead to the emaciation of the body.
- 112 When there is suppuration happening in the body, there will be interaction between the white blood cells with pathogens in the body. This mechanism is responsible for the manifestation of fever.

निद्रानाशः क्षवस्तम्भो गात्राणां रौक्ष्यमेव च॥८॥

शिरोहृद्गात्ररुग्ब्रूवैरस्यं गाढविट्कता।

शूलाध्माने जृम्भणं च भवन्त्यनिलजे ज्वरे॥९॥

Ma.Ni. - 2/8-9

- Involuntary movements.
- Irregularity in appearance, progression and remission of symptoms of *Jvara*.
- Dryness of throat, lips.
- Loss of sleep.
- Sneezing.
- Stiffness of the body parts.
- Dryness of the body parts.
- Headache.
- Chest discomfort.
- Bodyache.
- Abnormal taste perception.
- Passing of very hard faecal matter.
- Abdominal pain.
- Distention of the abdomen.
- Excessive yawning.

Pittaja Jvara:

वेगस्तीक्ष्णोऽतिसारश्च निद्राल्पत्वं तथा वमिः।

कण्ठौष्ठमुखनासानां पाकः स्वेदश्च जायते॥१०॥

प्रलापो वक्रकटुता मूर्च्छा दाहो मदस्तृषा।

पीतविण्मूत्रनेत्रत्वं पैत्तिके भ्रम एव च॥११॥

Ma.Ni. - 2/10-11

- Rapid onset, progression of the *Jvara*.
- Diarrhoea.
- Less amount of sleep.
- Vomiting.
- Suppuration in the throat.
- Pus formation in the lips.
- Pus formation in the oral cavity.
- Pus formation in the nasal region.
- Excess sweating.
- Irrelevant speech.
- Pungent taste in the mouth.
- Altered state of consciousness.
- Burning sensation in different body parts.
- State of intoxication.
- Thirst.
- Yellowish discolouration of urine.
- Yellowish discolouration of fecal matter.
- Yellowish discolouration of eyes.
- Giddiness.

Kaphaja Jvara:

स्तैमित्यं स्तिमितो वेग आलस्यं मधुरास्यता।

शुक्लमूत्रपुरीषत्वं स्तम्भस्तृप्तिरथापि च॥१२॥

(नात्युष्णागात्रता छर्दिरङ्गसादोऽविपाकितता)

गौरवं शीतमुक्केदो रोमहर्षोऽतिनिद्रता।

डम्बोतरोरुधो गल्पत्वं प्रसेको लवणास्यता।

नात्युष्णा गात्रता चर्दिराला लायोऽविपाकितता॥

प्रतिश्यायोऽरुचिः कासः कफजोऽणोश्च शुक्लता॥१३॥

Ma.Ni. - 2/12-13

- Feeling of covering of the body with wet clothes.
- Laziness.
- Sweetish taste in the mouth.
- Whitish discolouration of urine.
- Whitish discolouration of fecal matter.
- Stiffness of the body.
- Sensation of early satiety.
- Not too hotness on touching the body parts.
- Vomiting.
- Bodyache.
- Impaired digestion.
- Heaviness of the body parts.
- Sensation of coldness.
- Nausea.
- Horripilation.
- Excess sleep.
- Obstruction of the *Srotas* (channels).
- Mild discomfort.
- Excess salivation.
- Salt taste in the mouth.
- Rhinitis.
- Tastelessness.
- Cough.
- Whitish discolouration of the eyes.

Vatapittaja Jvara:

तृष्णा मूर्च्छो भ्रमो दाहः स्वप्ननाशः शिरोरुजा।

कण्ठास्यशोषो वमधू रोमहर्षोऽरुचिस्तमः॥१४॥

पर्वभेदश्च जृम्भा च वातपित्तज्वराकृतिः।

Ma.Ni. - 2/14-15

- Trishna – thirst.
- Murcha – altered state of consciousness.
- Bhrama – giddiness.
- Daha – burning sensation.
- Svapna Nasha – loss of sleep.
- Shiro ruja – head ache.
- Kantha shosha – dryness of the throat.

- Asya shosha – dryness of mouth.
- Vamathu – vomiting.
- Romaharsha – horripilation.
- Aruchi – tastelessness.
- Tama – darkness in front of the eyes.
- Parvabheda – tearing type of pain in the joints.
- Jrimbha – yawning.

Vatakaphaja Jvara:

स्तेमित्यं पर्वणां भेदो निद्रा गौरवमेव च॥15॥

शिरोग्रहः प्रतिश्यायः कासः स्वेदाप्रवर्तनम्।

संतापो मध्यवेगश्च वातश्लेष्मज्वराकृतिः॥16॥ Ma.Ni. – 2/15-16

- Staimitya – feeling of covering of the body with wet clothes.
- Parva bheda – tearing type of pain in the joints.
- Nidra – excess sleep.
- Gourava – heaviness of the body parts.
- Shiro Graha – stiffness of the head region.
- Pratishyaya – rhinitis.
- Kasa – cough.
- Sveda Apravartana – lack of sweating..
- Santapa – raised body temperature
- Madhyavega – moderate elevation of the symptoms neither too severe nor mild.

Pittakaphaja Jvara:

लिप्तातितास्यता तन्द्रा मोहः कासोऽरुचिस्तृषा।

मुहुर्दाहो मुहुः शीतं श्लेष्मपित्तज्वराकृतिः॥17॥

Ma.Ni. – 2/17

- Lipta Tiktasyata – coating of the tongue and bitter taste in the mouth.
- Tandra – drowsiness.
- Moha – altered state of consciousness.
- Kasa – cough.

- Aruchi – tastelessness.
- Trisha – thirst.
- Muhurdaha and Muhurshita – frequent feeling of burning sensation and coldness.

Sannipataja Jvara: the clinical features of Sannipataja Jvara is explained earlier in the present chapter.

Jvaramukta Lakshana:

स्वेदो लघुत्वं शिरसः कण्डूः पाको मुखस्य च।

क्षवथुश्चात्रलिप्सा च ज्वरमुक्तस्य लक्षणम्॥75॥ Ma.Ni. – 2/75

विगतक्लमसंतापमव्यथं विमलेन्द्रियम्।

युक्तं प्रकृतिसत्त्वेन विद्यात् पुरुषमज्वरम्॥329॥ Ca.Ci. – 3/329

देहो लघुर्व्यपगतक्लममोहतापः पाको मुखे करणसौष्टवमव्यथत्वम्।

स्वेदः क्षवः प्रकृतियोगिमनोऽत्रलिप्सा कण्डूश्च मूर्ध्नि विगतज्वर लक्षणानि॥

Atanka Darpana commentary, Ma.Ni. – 2/75

- Sveda – sweating¹¹³.
- Laghutva – lightness¹¹⁴ of the body.
- Shirasah Kandu – itching in the head region¹¹⁵.
- Mukhasya Paka – suppuration¹¹⁶ in the oral region.
- Kshavathu – sneezing.
- Annalipsa – liking towards the food.
- Vigataklama¹¹⁷ – disappearance of the fatigue.
- Vigata Santapa¹¹⁸ – subsidence of temperature.
- Avyatha¹¹⁹ – disappearance of discomfort.
- Vimalendriya¹²⁰ – proper functioning of Indriya.
- Prakriti Satva – proper functioning of Satva.

Upadrava of Jvara: Most of the Lakshana mentioned in the Upadrava of Jvara are related to Pranavaha Srotas, Rasavaha Srotas, Annavaha Srotas, Purishavaha Srotas. In advanced stages of Jvara, we are going to observe excess amount of fluid loss from the body, leading to the affliction of Udakavaha Srotas and in most of the Upadrava mentioned below we are going to observe the involvement of above said Srotas.

- 113 This is attributable to the deactivation of sympathetic nervous system which leads to peripheral vaso dilatation. The final outcome in this situation is excess sweating.
- 114 This is attributable to the subsidence of Ama in the body. This phenomenon is usually associated with appearance of appetite.
- 115 This is the specific Lakshana observed in the progression of the disease Jvara only. Whole body is having the dominance of Pitta Dosha at the time of the progression of the disease and when the disease is subsided, Pitta will be in diminished state, this in turn leads to the increase of Kapha Dosha, that too more prominently seen in Shira (head) as it is one of the Kapha Sthana and no other Kapha Sthana are involved. This is the peculiar feature of Jvara.
- 116 This is attributable to the increase of Pitta because of increase of Ushna quality in the body. This Lakshana is not seen at the beginning and it is considered to be peculiar feature of Jvara. This Lakshana is typically observed in the person who is suffering from Tridosha dominant Antarvega Jvara associated with Daha (burning sensation).
- 117 This is attributable to the peripheral vaso dilatation so that even the skeletal muscles receive adequate amount of blood supply and thus the person may have disappearance of fatigue.
- 118 This is attributable to the reduction of vitiated Pitta in the body.
- 119 All the main features of Jvara, i.e. Svedavarodha (lack of sweating), Santapa (burning sensation) and Sarvanga Grahana (stiffness of the body parts) will be subsided. There will be subsidence of vitiated Pitta too.
- 120 This is attributable to peripheral vaso dilatation. This is occurring not only in the skin and even occurs in different sense organs too. This phenomenon is responsible for the proper functioning of Indriya. For example, Shrotrendriya, proper restoration of formation of perilymph and endolymph is responsible for proper hearing.

'हिवकाभासकासतृष्णाच्छर्यतीसारमूर्च्छाङ्गभेदारोचकाः कुक्ष्यवि- कता चेति दशोपद्रवा ज्वरमुपद्रवन्ति' इत्युक्ताः॥- Chakra, Ca.Ci. - 3/49

- **Hikka** – hiccup¹²¹.
- **Shvasa** – breathing discomfort.
- **Kasa** – cough.
- **Trishna**¹²² – thirst.
- **Chardi** – vomiting¹²³.
- **Atisara** – diarrhoea.
- **Murcha**¹²⁴ – altered state of consciousness.
- **Angabheda**¹²⁵ – bodyache or myalgia.
- **Arochaka**¹²⁶ – tastelessness.
- **Krichra Vitkata**¹²⁷ – difficulty in passing the bowels.

Asadhya Lakshana of Jvara:

हेतुभिर्बहुभिर्जातो बलिभिर्बहुलक्षणः॥50॥

ज्वरः प्राणान्तकृद्यश्च शीघ्रमिन्द्रियनाशनः।

Ca.Ci. - 3/50-51

Jvara is considered to be life threatening in the following conditions. The description of the points mentioned are explained elsewhere in the current chapter.

- If the *Jvara* is manifested due to multiple and virulent aetiological factors¹²⁸.
- If *Jvara* is associated with multiple symptoms¹²⁹.
- If *Jvara* is associated with loss of function of *Indriya*¹³⁰ in short time.

Key summary of the chapter

स्वेदावरोधः सन्तापः सर्वाङ्गग्रहणं तथा॥13॥ विकारा युगपद्यस्मिन् ज्वरः स परिकीर्तितः।

Su.Ut. - 39/13-14

- Absence of sweating, raised body temperature and stiffness of the body parts are observed either in isolation or altogether in an individual at a time is termed as *Jvara*. Several definitions are available in the literatures as far as the definitions of *Jvara* are concerned.
- *Pitta Dosha* is the primary *Dosha* afflicted in *Jvara*, *Rasa Dhatu* will be primarily involved. Naturally, we can infer the involvement of *Rasavaha Srotas* with *Sanga* and *Vimargagamana* as the type of *Srotodushti* during the progression of *Jvara*.

- 121 There is a possibility of involvement of *Pranavaha Srotas* along with *Rasavaha Srotas* affliction. When an individual is presented with *Upadrava* it may be related to same *Srotas* or nearby *Srotas*.
- 122 This is attributable to affliction of *Udakavaha Srotas* as there is excess *Udaka Kshaya* (fluid loss) in due course of time because of either *Atimutrata* (increased micturition) or *Svedadhikya* (excess sweating).
- 123 If the person is suffering from fever associated with either vomiting or diarrhoea or both, there will be excess loss of water and electrolytes including sodium, potassium, calcium ions. If the condition persists for more time, the person may develop serious complications very rapidly sometimes within half to one hour.
- 124 This can be attributable to reduced nutrient absorption from the alimentary canal including water and electrolytes and there will be poor blood supply to the brain due to hypovolemia.
- 125 When an individual is presented with myalgia, it indicates increased muscular activity which in turn leads to increased heat production. In other words, the pathogenesis of fever is not checked.
- 126 This can be attributable to the involvement of *Prahavaha Srotas*. This may occur as a result of poor gastrointestinal secretions or abnormal autonomic nervous system activity. Increased sympathetic activity will lead to poor oral and gastrointestinal secretion and parasympathetic nervous system stimulation will lead to enhanced oral and gastrointestinal secretion. Thus, in both occasions, the person may present with tastelessness.
- 127 This is attributable to affliction of *Purishavaha Srotas* and this *Srotas* is a continuation of *Mahasrotas* which comprises *Pranavaha*, *Annavaha* and *Purishavaha Srotas* in the sequential order. This may occur as a result of poor gastrointestinal motility and hampered gastrointestinal secretion and contraction of sphincters for most of the time during progression of fever.
- 128 Let us learn this concept with one example hypothetically. Suppose we have come across with three individuals A, B and C suffering from *Kaphaja Jvara*. Person A is having the consumption of curds for 2 days, person B is having the consumption of curds, banana for 5 days and person C is having the consumption of curds, ice cream and living in air condition room since 20 days. In all these occasions, the vitiation of *Kapha Dosha* is varied. In person A, there will be mild vitiation of *Kapha Dosha*, in person B, there will be moderate vitiation of *Kapha Dosha* and in person C, there will be severe vitiation of *Kapha Dosha* in the individuals. Accordingly, as far as management of *Kaphaja Jvara* is concerned, treatment of *Kaphaja Jvara* is becoming easier in person A when compared to B and C, and in person B, treatment is easier when compared to *Kaphaja Jvara* in person C, but difficult to manage when compared to person A.
- 129 Let us discuss the *Lakshana* mentioned in the definition of *Jvara* and their analysis in assessment of prognosis of the disease. *Svedavarodha*, *Santapa* and *Sarvangagrahana* are mentioned in the definition of *Jvara*. When one observes only *Santapa*, but not *Svedavarodha* and *Sarvangagrahana*, it indicates either localised pathogenesis or milder pathogenesis or low virulence of microorganisms. When an individual is having *Svedavarodha* along with *Santapa*, it indicates peripheral vasoconstriction, it indicates the moderate pathological events in the suffering individual. If an individual is showing all the *Lakshana* observed in an individual, that indicates severe form of pathological events or high virulence of microorganisms etc. Suppose, if the individual is presented with above said symptoms, associated with cough or pain abdomen etc, then it indicates wide spread involvement of bodily organs suggests the much poor prognosis of the disease.
- 130 When an individual is presented with loss of function of *Indriya*, it indicates hypovolemia where *Indriya* which are mostly seen in the periphery will receive minimum blood supply to channelise more blood to the core of the body – brain and heart. Thus, the person may develop hypovolemic shock or hypoglycemia or hyponatremia or dehydration.

- Vishesha Purvarupa is the special feature of Jvara which indicates the probable Dosha involved in the forthcoming type of Jvara. Atyarthha Jrimbha (excess yawning) is seen as specific Purvarupa of Vataja Jvara, Nayana Daha (burning sensation around orbital region) is the specific Purvarupa of Pittaja Jvara and Anannabhinandana (lack of appetite) is the specific Purvarupa of Kaphaja Jvara.
- Vidhi Samprapti is special category of classification of Jvara or any type of disease, as it helps proper management plan of Jvara.
 - 2 types of Jvara – Sharirika and Manasika
 - 2 types of Jvara – Soumya and Agneya
 - 2 types of Jvara – Antarvegi and Bahirvegi
 - 2 types of Jvara – Sadhya and Asadhya
 - 2 types of Jvara – Ushnabhiprayaja and Shitabhiprayaja
 - 2 types of Jvara – Prakrita and Vaikrita
 - 3 types of Jvara – Ama, Pachyamana and Nirama Jvara
 - 5 types of Vishama Jvara – Santata, Satata, Anyedyushka, Tritiyaka, Chaturthaka
 - 7 types of Dhatugata Jvara – Rasadhatugata, Raktadhatugata, Mamsadhatugata, Medodhatugata, Asthidhatugata, Majjadhatugata and Shukradhatugata
 - 8 types of Jvara – Vataja, Pittaja, Kaphaja, Vatapittaja, Pittakaphaja, Vatakaphaja, Sannipataja, Agantuja
 - 13 types of Sannipataja Jvara
 - 4 types of Agantuja Jvara – Abhighataja, Abhishangaja, Abhicharaja, Abhishapaja
 - Punaravartaka Jvara – depending upon recurrence of Jvara
- Karnamulika Shotha (swelling of parotid gland) is the specific type of Upadrava seen in the patients of Sannipataja Jvara.
- Disturbances in heat regulation either increased heat production in the body or hampered heat loss from the body or both of the mechanisms are responsible for the manifestation of fever.
- **Mechanisms of heat generation:**
 - Increased basal rates of metabolisms of all the cells of the body
 - Extra rate of metabolism caused by muscular activity including muscular contraction caused by shivering
 - Extra metabolism caused by the effect of thyroxine on the cells
 - Extra metabolism caused by the effect of epinephrine, norepinephrine and sympathetic stimulation on the cells
 - Extra metabolism caused by increased chemical activity in the cells themselves, especially when the cell temperature increases
 - Extra metabolism needed for digestion, absorption and storage of food
- **Mechanisms of heat loss:**
 - Slower conduction of heat from the site of production of heat in the core to the skin
 - Slower conduction of heat from the skin to the surroundings

Self-Assessment: Questions for Reviews

Long essays

- Define Jvara and explain Amajvara and Nirama Jvara Lakshana
- Write Nidana, Samprapti and Bheda of Jvara. Explain Vishama Jvara
- Explain Jvara Samprapti and types of Vishama Jvara
- Discuss Sannipata Jvara
- Define Jvara and enumerate the Vidhi Samprapti of Jvara

Short essays

- Amaja and Nirama Jvara Lakshana
- Explain Sannipataja Jvara
- Describe Samanya Samprapti of Jvara
- Describe Rasa, Rakta and Mamsa Ashrita Jvara Lakshana
- Write about Jvaramukta Lakshana
- Describe Pachyamana Jvara Lakshana
- Describe Sannipataja Jvara Upadrava

- ⊗ Dhatugata Jvara
- ⊗ Describe Meda, Asthi and Majja Ashrita Jvara Lakshana

MCQs

1. Pradhana Dosha in the manifestation of Jvara is

- a. Vata; b. Pitta;
- c. Kapha; d. Rakta

Answer: (b)

2. The Dosha dominance observed in Soumya Jvara is

- a. Vata;
- b. Kapha;
- c. Vata and/or Kapha;
- d. Vata and Pitta

Answer: (c)

3. Which of the following Prakrita Jvara is considered as Krichrasadhya (difficult to cure)?

- a. Prakrita Vataja Jvara;
- b. Prakrita Pittaja Jvara;
- c. Prakrita Kaphaja Jvara;
- d. All of the above

Answer: (a)

4. Karnamulika Shotha is the Upadrava of

- a. Vataja Jvara;
- b. Sannipataja Jvara;
- c. Antarvegaja Jvara;
- d. Prakrita Vataja Jvara

Answer: (b)

5. Nayana Daha is the Vishishta Purvarupa of Jvara.

- a. Soumya Jvara;
- b. Pittaja Jvara;
- c. Santata Jvara;
- d. Rasadhatugata Jvara

Answer: (b)

6. Dhatu involved in the manifestation of Tritiyaka Jvara is

- a. Rasa; b. Meda;
- c. Rakta; d. Shukra

Answer: (b)

7. The main affliction in Abhigataja Jvara is

- a. Vata Vriddhi and Rakta Kshaya;
- b. Vata and Pitta vitiation;
- c. Sannipata vitiation;
- d. Rakta Kshaya only

Answer: (a)

8. Sankhya Samprapti of Jvara is

- a. 2; b. 3;
- c. 5; d. 8

Answer: (d)

9. Bahumutrata (excess micturition) is the Lakshana seen in

- a. Ama Jvara;
- b. Santata Jvara;
- c. Rasadhatugata Jvara;
- d. Soumya Jvara

Answer: (a)

10. As per Madhavakara, types of Tritiyaka Jvara are

- a. 2; b. 3;
- c. 4; d. 5

Answer: (b)

11. Adhishtana of Jvara is

- a. Sharira;
- b. Manas;
- c. Both Sharira and Manas;
- d. None of the above

Answer: (c)

12. Bahirvega Jvara is

- a. Asadhya;
- b. Krichra Sadhya;
- c. Sukhasadhya;
- d. Unpredictable

Answer: (c)

13. The Sukhasadhya Lakshana of Jvara includes

- a. Tulya Dushyata;
- b. Jirnata;
- c. Tulya Rtu Doshata;
- d. Tulya Dosha Dushya

Answer: (c)

14. Abhishangaja is a type of

- a. Doshaja Jvara;
- b. Dhatugata Jvara;
- c. Sannipataja Jvara;
- d. Agantuja Jvara

Answer: (d)

15. Asveda is the Lakshana of

- a. Bahirvegi Jvara;
- b. Antarvegi Jvara;
- c. Dhatugata Jvara;
- d. Pralepaka Jvara

Answer: (b)

16. The period of Vataja Sannipataja Jvara is

- 7 days;
- 10 days;
- 12 days;
- Innumerable

Answer: (a)

17. Vishamo Vishamarambho Kriyakalo Anushangavan is said by

- Charaka;
- Vagbhata;
- Sushruta;
- Sharangadhara

Answer: (b)

18. Vomiting and diarrhoea both symptoms are seen in

- Raktagata Jvara;
- Majjagata Jvara;
- Asthigata Jvara;
- Medogata Jvara

Answer: (c)

19. Jvara is considered as

- Kalaja Roga;
- Aupasargika Roga;
- Svabhavika Roga;
- Sahaja Roga

Answer: (b)

20. Jrimbha is considered as of Vataja Jvara.

- Samanya Lakshana;
- Samanya Purvarupa;
- Vishishta Purvarupa;
- Upadrava

Answer: (c)

21. Types of Vishama Jvara are

- 3;
- 2;
- 7;
- 5

Answer: (d)

22. Shukradhatugata Jvara is considered as

- Sukhasadhya;
- Krichrasadhya; Yasya;
- Asadhya

Answer: (d)

23. The Dhatu involvement seen in the patient of Anyedyushka Jvara is

- Rakta;
- Mamsa;
- Medas;
- Asthi and Majja

Answer: (b)

24. The Dosha dominance observed in the patients of Vata-balasaka Jvara is

- Vata and Pitta;

- Vata and Kapha;
- Pitta and Kapha;
- Vata, Pitta and Kapha

Answer: (b)

25. Which of the following is true in the case of Pittaja Jvara?

- Pittaja Jvara in Sharat Rtu is considered as Prakrita Jvara
- Pittaja Jvara in Vasanta Rtu is considered as Vaikrita Jvara
- Pittaja Jvara in Sharat Rtu is considered to be Sukhasadhya in nature.
 - a is correct
 - a and b are correct
 - a, b and c are correct
 - none of the above statements are correct.

Answer: (iii)

Chapter resources

- <https://www.zuniv.net/physiology/book/chapter21.html>
- Charaka Samhita, Nidana Sthana – 1st chapter – Jvara Nidana Adhyaya
- Charaka Samhita Chikitsa Sthana – 3rd chapter – Jvara Chikitsa Adhyaya
- Sushruta Samhita, Uttara Tantra – 39th chapter – Jvara Pratishedha Adhyaya
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
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